

Prices vs. Production Restrictions: China's Coal Industry*

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Abstract

Governments often use command-and-control production restrictions, such as plant closures or mandated output cuts, to manage externalities, when setting up a cap-and-trade system is difficult or time-consuming. Unlike an efficient tax or quota, these restrictions need not reduce output in order of producer efficiency and can create substantial misallocation. We study China's 2016 coal reform, which closed small mines and curtailed output at large mines to support industry profitability and employment while reducing pollution and mining accidents. Using an empirical model of interprovincial coal demand and supply, we show that the observed restrictions were distortionary, but can be justified relative to a tax when the government places greater weight on producer surplus.

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1 Introduction

Regulators often use price- and quantity-based policies to mitigate externalities. Economists have long advocated for market-based price or quantity instruments, such as Pigouvian taxes (Pigou, 1920; Baumol, 1972) or cap-and-trade systems (Crocker, 1966; Dales, 1968; Montgomery, 1972), to internalize externalities efficiently. Although resource taxation is common, sophisticated cap-and-trade markets are often infeasible due to high transaction costs (Nie, 2012) or weak institutional capacity (Bell and Russell, 2002; Blackman and Harrington, 2000), especially in developing countries. In these cases, the practical policy choice is between a tax and a command-and-control restriction on production, such as factory shutdowns or moratoria on resource extraction (Blackman, Li and Liu, 2018).

We study a major reform of China’s coal industry launched in 2016. Policy choices in this setting are consequential. As the world’s largest producer and consumer of coal, China faces acute tensions between energy security, environmental consequences, and economic stability. It has been widely reported that the reliance on coal undermines China’s energy transition and global efforts to mitigate climate change (McGrath, 2019; WSJ, 2022; Bloomberg, 2022). A number of papers have documented the severe air-quality and health consequences associated with coal (Almond et al., 2009; Chen et al., 2013; Ebenstein et al., 2017). At the same time, coal mining provides significant employment opportunities and local government revenues, especially in major producing provinces dominated by state-owned enterprises (Wright, 2012; Hervé-Mignucci et al., 2015). Policies affecting coal output therefore have large economic and distributional consequences (Clark and Zhang, 2022; Yuan et al., 2025).

As part of a multi-year effort to reduce coal consumption and pollution while preserving employment and profitability of the industry, the Chinese government in April 2016 ordered small mines to close and large mines to reduce output. As provinces implemented these measures, coal prices surged: our data show that the average coal price in the second half of 2016 rose by 31.0% to RMB 539 (\$77.4)/ton, from RMB 411 (\$60)/ton in the same period of 2015. The government continued to close mines and strictly monitor production to reduce production above the approved capacity at least through 2018.

From a welfare perspective, why might the government prefer this policy to taxation? We first show that it is theoretically ambiguous whether a tax or a production restriction achieves higher social welfare. Following Weitzman (1974), price and quantity instruments can be ranked differently under uncertainty depending on the underlying slopes of marginal benefits and marginal costs. In our setting, shocks to demand, marginal costs, and externalities all matter for this comparison. Furthermore, we show how the preferred policy shifts when policymakers place greater weight on certain objectives, such as firm profits rather than

consumer surplus.

To analyze the demand and supply in the coal market, we combine the policy intervention with a comprehensive dataset on coal production and consumption. Our main data set includes monthly prices at the province level and shipment quantities from producers to end users both within and across provinces. We validate the shipment data with the production and consumption data from other sources.

We next formulate a model that incorporates key institutional features of China’s massive and partially regulated coal market. First, coal procurement in the electricity and other sectors has undergone significant market reforms since 2013, essentially allowing the market to set prices that equilibrate demand with supply. Second, the shares of coal shipped from one province to another are highly stable over time, despite varying price dispersions across provinces. Third, large coal producers and buyers often operate under long-term quantity contracts. These contracts at the time often promised a delivery quantity but allowed prices to adjust based on spot market prices. In our model, the market is competitive. We specify a demand function and a supply function for each province. The local price in each province is determined by the province’s demand and the total supply from different provinces. The quantity supplied by a province is determined by a weighted average of prices in provinces it sells to. The effects of the policy are modeled as province-specific shifts of the supply functions.

We estimate the demand and supply of coal across provinces. To identify demand, we construct price instruments using variations in the policy start dates and policy compliance levels across provinces that shift supply. The key identifying assumption is that the policy exclusively affects supply and is not correlated with the demand unobservables. We also consider a variety of alternative instruments. Similarly, we estimate supply using demand shifters. Our preferred threshold estimates suggest that the policy significantly steepens aggregate marginal costs once production approaches capacity.

Given the estimated equilibrium model of coal demand and supply, we conduct a series of counterfactual simulations to quantify the effects of enforcing the production restriction. First, we find that removing the restrictions would increase the annual production by more than 693 million tons and decrease the price by more than RMB 161/ton. Using reasonable estimates of coal’s environmental impact, we find that the loss of economic surplus is offset by the saved environmental costs.

Second, we compare the policy with an efficient quota or a cap-and-trade system to understand its distortionary effects. Our supply estimates suggest that the policy increases aggregate marginal costs especially near approved capacity, and that larger, lower-cost provinces face stronger post-threshold cost increases. These findings suggest inefficient reductions of

quantities both within and across provinces. Holding fixed each province’s observed quantity, removing the within-province distortion would raise producer surplus by RMB 156 billion, and efficiently reallocating quantity reductions across provinces would further raise producer surplus by RMB 90 billion.

We then use the estimates to rank the production restriction policy against a tax. Given a welfare function that consists of consumer and producer surpluses, tax revenue, and avoided pollution damages, we find that social welfare would be RMB 266 billion higher under an optimal tax. We also show that this result is not sensitive to including an additional external loss as a function of total quantity in the welfare function. This external loss can reflect social cost of carbon, but more generally captures any social costs that we do not quantify but rise with aggregate production. We show that the optimal tax can achieve a higher welfare value for any such increasing loss function.

Finally, we rationalize the policy choice with heterogeneous policy preferences for producer surplus and avoided pollution damages. We find that valuing the producer surplus 37% more than other components in the welfare objective function alone would allow the observed policy to achieve a higher objective value than any tax. The intuition is that the weighted loss of producer surplus under a tax can be so large that tax revenue cannot offset it, whereas production restrictions can directly increase producer surplus. We also find that a greater weight for avoided pollution damages must be paired with an even greater preference for producer surplus for the government to prefer the observed policy to a tax.

Related Literature and Contribution

We first contribute to a literature that uses observed policy choices to understand the political or social preferences. In trade, Goldberg and Maggi (1999) estimate the government’s relative weight on aggregate welfare versus political contributions. Subsequent work examines related political-economy models of trade protection (Gawande and Bandyopadhyay, 2000; Mitra, Thomakos and Ulubaşoğlu, 2002; McCalman, 2004; Matschke and Sherlund, 2006; Gawande, Krishna and Olarreaga, 2009; Francois and Nelson, 2014; Adão et al., 2026). Related work estimates social welfare weights that rationalize observed tax-benefit systems (Bourguignon and Spadaro, 2012; Bargain et al., 2014; Hendren, 2020) and infers political motives from infrastructure choices, including rail in California and highways in China (Fajgelbaum et al., 2024; Alder and Kondo, 2019). Relative to this literature, which typically infers preferences from variation within a given policy framework, we use the government’s choice between instruments, the observed production restriction vs. tax, to bound the relative weights on components of the welfare objective function.

Our paper also contributes to the empirical literature on measuring the economic effects of environmental regulations.¹ A number of papers focus on productivity impacts (e.g., Jaffe et al., 1995; Berman and Bui, 2001; Greenstone, 2002; Greenstone, List and Syverson, 2012; He, Wang and Zhang, 2020). Using estimated equilibrium models, Ryan (2012) and Fowlie, Reguant and Ryan (2016) emphasize the roles of market power. Chen et al. (2025) study how conglomerates reallocate internal production. In this paper, we develop and estimate a new competitive equilibrium model for China’s coal sector.

We also note that there have been few equilibrium analyses of China’s coal industry in economics despite its enormous importance to China’s economy and the global environment. One reason might be the lack of granular and recent data.² For example, Zhou et al. (2019) study the evolution of coal mining firms’ productivity, and Zheng (2024) studies the effects of buyer market power on productivity and safety, both using a production function approach based on manufacturing census data from before 2007. In our paper, we combine a detailed dataset, the institutional knowledge of the market, and the industry reform to address the endogeneity problems in the estimation of coal demand and supply in our equilibrium model. Our estimates of demand and supply elasticities may be of independent interest for other researchers studying China’s coal industry.

Finally, we build on the classic framework of Weitzman (1974)³ to compare taxation with production restriction. Weitzman (1974) studies the choice between price and quantity instruments under uncertainty, while subsequent work compares taxes with permits, standards, command-and-control rules, and hybrids (Adar and Griffin, 1976; Roberts and Spence, 1976; Koenig, 1985; Helfand, 1991; Dickie and Trandel, 1996). Our comparison differs in three respects. First, demand uncertainty, which is often absent in a local comparison, matters for the instrument ranking because the restriction changes the effective supply schedule instead of fixing a quantity, so output and welfare remain responsive to demand shocks. Prior papers find that demand shocks play a role under different demand or supply systems (Kolstad,

¹More broadly, we note that there is a growing literature on the empirical evaluation of industrial policy (Kline and Moretti, 2014; Aghion et al., 2015; Alder, Shao and Zilibotti, 2016; Juhász, 2018; Kalouptsi, 2018; Lane, 2018; Miravete, Moral and Thurk, 2018; Lashkaripour and Lugovskyy, 2018; Criscuolo et al., 2019; Rotemberg, 2019; Barwick, Kalouptsi and Zahur, 2019; Giorcelli, 2019; Yi, Lawell and Thome, 2019; Hanlon, 2020; Bai et al., 2020; Fan and Zou, 2021; Giorcelli and Li, 2021; Guo and Xiao, 2022; Aldy, Gerarden and Sweeney, 2022).

²Existing studies that estimate or calibrate the coal demand or supply typically use aggregate annual data (see, for example, Burke and Liao, 2015; Shi, Rioux and Galkin, 2018; Teng, Burke and Liao, 2019). A number of papers have studied the coal-mining industry in other countries, with a focus on safety and productivity (Sider, 1983; Gowrisankaran et al., 2015), technology adoption (Rubens, 2022), labor market competition (Delabastita and Rubens, 2022; Demirer and Rubens, 2025), and the decline of the US coal industry (Watson, Lange and Linn, 2023).

³For the large literature related to Weitzman’s framework, see, for example, Williams III (2002), Hepburn (2006), Weitzman (2020), and Stavins and Wagner (2022).

1986; Stavins, 1996), or for attribute-based regulation (Kellogg, 2020). Second, we further relate the welfare comparison to policy preferences and show how they interact. Third, our counterfactual analysis simulates the market equilibrium under alternative policies, going beyond a local approximation. Our approach based on estimated demand and supply models also differs from prior work comparing policy instruments, which uses integrated assessment models (Pizer, 2002; Muller and Mendelsohn, 2009) or calibrated theoretical models (Newell and Pizer, 2003; Fell, MacKenzie and Pizer, 2012).

In the rest of the paper, we first describe the main institutional features of China’s coal market in Section 2. Then in Section 3, we illustrate the welfare comparison between tax and production restriction in a conceptual model. Section 4 lists the various datasets used in the analysis. Sections 5 and 6 present the demand and supply models and their estimation. We use the model to analyze the reform of China’s coal industry and how the observed policy choice reveals policy preferences in Section 7. Section 8 concludes.

2 Empirical Background

2.1 Coal Types, Quality and Mining Technology in China⁴

Over 70% of coal produced in China is bituminous coal, and 23% is anthracite.⁵ High-purity bituminous coal and about half of anthracite are used for coking, which in turn produces coke for steel-making. The rest of these two types of coal are mainly used by power plants for electricity production. The main uses of lower-purity coal are for construction (cement making) and heating.

Coal purity and extraction methods vary significantly across regions. Shanxi and Inner Mongolia have the largest reserves and are the largest coal-producing provinces, accounting for over a quarter of national output.⁶ Shanxi also produces the highest-quality coal, with an average heat content (defined as the total energy produced per kilogram after complete combustion) of about 6,242 kcal/kg, compared to the national average of 5,350 kcal/kg and the US average of 5,600 kcal/kg. Over 90% of mines in China are pithead mines, where coal is extracted from deep underground. The average mine depth is 456 meters. This extraction method is more costly than open-pit mining more commonly found in countries such as Australia and India, and regions such as the western US.

⁴This section is based on Nathaniel Aden, David Fridley and Nina Zheng (2009) and National Energy Report (2020).

⁵Coal is usually classified by the carbon content (by weight). The carbon content is over 86% for anthracite and between 45% and 86% for bituminous coal. The remaining coal falls into the categories with lower carbon content, such as subbituminous, lignite and brown coal (American Geosciences Institute, n.d.).

⁶Appendix Figure I.2 shows the distribution of coal reserves.

2.2 The Formation of the Coal Market

Historically, the Chinese government exercised tight control over both the supply and demand of coal through planned prices and quantities. The government partially liberalized the market in 1993 by allowing coal to be traded while maintaining special contracts that gave the large state-owned power plants the option to buy coal from mines at planned quantities and prices (Yang et al., 2018). The special contract prices were typically lower than the prevailing market prices. When contracted coal was insufficient to meet electricity demand, power plants turned to the spot market to purchase additional coal. The government further liberalized the market in a series of reforms, and fully abolished the special contracts in 2013 (State Council, 2013). These reforms also changed the procurement, operation, and investment decisions of power plants, which then started to respond to fluctuating coal prices in a manner more consistent with profit-maximizing firms (Xu and Chen, 2006; Ma, 2011; Liu, Margaritis and Zhang, 2013; Zhao and Ma, 2013; Gao and Van Biesebroeck, 2014; Eisenberg, 2019, 2024).

2.2.1 Method of Sales

Delivery Contracts. During our sample period, coal buyers and producers trade via both long-term (annual or multi-year) contracts and the spot market. Major buyers include coal power plants, municipal central heating systems,⁷ construction firms, steel makers, and fertilizer manufacturers. A long-term contract is signed directly between a coal producer and an end user (not a trading intermediary). Provincial governments are often involved to help firms find their matches, especially in large coal-producing provinces, which bear political responsibility to ensure coal supply to other provinces' power plants and heating systems (NDRC, 2021b). These contracts would nominally promise an annual delivery quantity but allow prices to adjust on a quarterly or even monthly basis, based on the spot market prices (e.g., Shanghai Securities News, 2014, 2015; ICBC, 2016). As a result, actual delivered quantities may deviate from the contractual terms (International Energy Network, 2016a).

Monthly Price Discovery. A standard practice is for long-term contract buyers and sellers to use public monthly price indices to formulate a price. The most commonly used indices during and after our study period are average transaction prices (net of transportation costs) of coal sold to power plants in different regions (NDRC, 2016c, 2017a). Specifically, buyers

⁷Across northern China, most residential apartments, government facilities, and schools receive heating through municipal centralized systems that primarily burn coal to generate heat. This infrastructure consists of large coal-fired boilers that heat water, which is then distributed through an extensive network of insulated pipes to various buildings (Chen et al., 2013).

and sellers update their contract prices based on the signing prices (which may depend on coal quality in addition to projected supply and demand) and the average delivery prices in the past month published by coal trade associations and port authorities, which may include trades based on the contracts and in the spot market.⁸

Contract Compliance. As mentioned above, a large number of news reports at the time suggested that almost none of the contracts were executed at the contracted price in 2014-2016. Specifically, reports (Li, 2014; Zhao, 2014, 2015) suggest that major coal miners were able to sell coal at the contract prices in the first two months of 2014 but had to reduce prices soon after market prices started to falter.⁹ The compliance level was equally poor in 2015 and 2016 (Bie, 2015; Wang, 2015).

Compliance significantly improved in 2017 as a result of three forces. First, the industry reform that we will describe below lifted the prices. Second, the new contracts specify a range of prices that become more practical to comply with when spot market fluctuate. Third, the government implemented additional measures in 2017 to penalize power plants for not buying the full amount of contracted coal, such as excluding non-compliant power plants from the wholesale electricity market (Gao, 2017). These measures limited the spread between the spot market and contract prices (China National Coal Association, 2018). Later reports (Zhang, Hong and Wu, 2020) suggest that over 2 billion tons of coal, or 50% of the 2017 quantity, were contract-compliant.

2.3 Economic Policies

2.3.1 Coal Supply Reform

In the five-year plan published by the National Development and Reform Commission (NDRC, China’s macroeconomic management agency) in 2016, the Chinese government outlined a plethora of goals and plans to restructure the coal-mining industry. Specifically, the gov-

⁸International Energy Network (2016*b*) provides an example on how to reset the monthly price using the average coal prices sold to power plants (BSPI) and the average price of coal for other uses (OPI) around the gulf of Bo Hai, both published by Qinhuangdao Ocean Shipping Coal Trading Market Co. Given the contract price p , the indices p_t^{BSPI} and p_t^{OPI} in month t , and the heat content of coal h measured in kcal/kg, the new price in month $t + 1$ is adjusted to $0.5p + (0.25p_t^{\text{BSPI}} + 0.25p_t^{\text{OPI}}) \frac{h}{5,500\text{kcal/kg}}$. The government advises buyers and sellers to explicitly specify this particular form of the formula (dynamic price updates linear in average coal prices) in their long-term contracts. Contract prices, the price indices and the assigned weights could differ across firms and across years.

⁹Hu (2014) reports that contracts at the beginning of 2014 covered 1.9 billion tons of coal. Our data suggest that the first two months’ quantity typically accounts for 15% of the annual level. A back-of-the-envelope calculation shows that the contract-compliant share from the early months in 2014 would be about 7% of the total coal quantity (4 billion tons).

ernment explicitly called for improving the industry’s profitability, limiting coal production, and reducing the industry’s environmental impact.¹⁰ Based on this plan, the State Council, China’s chief administrative body, issued orders in February 2016 to limit the number of working days for coal mines from 330 to 276 days. The order also called for the closure of small coal mines. We refer to these orders as the **supply reduction policy** in the paper.¹¹

The actual dates of implementation and compliance with the policy varied across provinces and coal mine ownership types. Shanxi, historically China’s largest coal-producing province, was the first to implement the policy in April 2016. Other leading coal-producing provinces such as Shaanxi and Inner Mongolia began their implementation in May. Large state-owned coal mines were also subject to more stringent monitoring, and their output fell more (Appendix I.2). Coal prices surged in the second half of 2016 relative to the prior months in 2016 and to the same period in 2015. NDRC officially suspended the 276-working-day policy in October amid concerns over meeting the (highly inelastic) winter heating and power demand (Lai, 2016; Shi, Rioux and Galkin, 2018). Nonetheless, mine closures and strict production monitoring continued in 2017 and 2018. The policy was gradually eased in 2019 (NDRC, 2019).

2.3.2 Resource Taxation

China instituted a resource tax on coal production in 2014. The central government established a band for the coal resource tax rate of 2 to 10% on the sales revenues of the raw coal extracted,¹² within which provincial governments choose the specific rate, subject to the central government’s approval (STA, 2014). Major provinces such as Shanxi set a rate at 8%, and others set lower rates (Qin, Zhang and Xie, 2020).

Policymakers did not use this tax as the main instrument in the coal industry reform two years later for several reasons. First, they designed the new tax to reduce firms’ tax burden and fund provincial spending on coal-production-related environmental damage as part of a broader fiscal reform (Xinhua, 2014), which replaced a patchwork of fees on coal production with a single levy. Raising the rate to restrict output would have clashed with that political narrative. Second, fiscal reform moves slowly, while the central government saw cutting coal capacity and output as an urgent task better handled through fast, targeted administrative mandates (Li, 2016; People’s Daily, 2016). Finally, the central government allows provincial

¹⁰In Appendix A, we provide snapshots of the original document and its translation.

¹¹The Chinese press often referred to this policy as “capacity reduction policy”, reflecting the closure of small mines. Given that the policy also reduced the output of operating mines, we use the term “supply reduction” as a compromise.

¹²This revenue levy does not additionally distinguish between different grades of the coal, but allows for reduced rates on coal extracted in mines near depletion.

governments to keep all resource tax revenues as an incentive to invest in environmental protection (Hu, 2017), making it hard to further redistribute the revenue directly to coal consumers or producers.

2.3.3 Emission Trading System

China’s carbon market for coal-fired power was introduced through regional emissions-trading pilots and then consolidated into the national Emissions Trading System (ETS) for the power sector. The institution was developed over a long time horizon: seven regional pilots began in 2013–14 and, after several years of operation, covered only about 7 percent of national CO₂ emissions (Almond and Zhang, 2021), while the national program did not begin trading until 2021 and initially applied only to the power sector (Goulder et al., 2022). The market adopts a tradable performance standard design, instead of cap-and-trade (Ministry of Ecology and Environment, 2020; Goulder et al., forthcoming): allowances are allocated using fuel- and technology-specific emissions-intensity benchmarks, and pre-allocation was tied to lagged generation, with final allocations adjusted to match the realized output.¹³ These design features incentivize power plants to reduce emissions intensity, rather than to reduce generation or accelerate coal-capacity retirement. As a result, both the institutional design and the long implementation horizon of the ETS were poorly aligned with the 2016 coal industry reform’s objective of reducing coal output.

2.3.4 Other Economic Policies

We also note two other sets of significant economic policies during our sample period. First, the government expanded credit and increased infrastructure spending to counter the decline in housing and stock market prices in 2015 and 2016 (Brandt et al., 2020). Many infrastructure projects, such as the national high-speed rail system, experienced rapid expansion in subsequent years (Reuters, 2016). Second, the government started to close small or unproductive plants in other industries.¹⁴ These closures reduced 10% of state-owned steel capacity in 2016 (Lu, 2016). However, we expect the plant closures to have a small effect on coal demand given that the output share of closed plants is likely low. Between 2016 and

¹³Allowances in 2019–2020 were first pre-allocated using 70 percent of 2018 electricity/heat supply, but after 2019–2020 emissions data were verified, the final quota was set according to the generator’s actual 2019–2020 electricity/heat supply; if the final approved quota differed from the pre-allocation, excess allowances were withdrawn or missing allowances were added.

¹⁴It is of separate interest that the government, as opposed to the market, needs to order the shutdown of the plants. A plausible explanation is that many plants are supported by the local governments to provide employment despite being unprofitable. To facilitate the shutdown of coal and steel plants, China’s Ministry of Finance created a fund of RMB 100 billion (\$15.3 billion) to provide relief to the laid-off employees (Lu, 2016).

2018, coal demand remained strong as the main consuming sectors—thermal power, coking, steel and cement—either expanded output or maintained historically high production levels, against the backdrop of continued economic growth.¹⁵

3 Conceptual Framework

We compare a tax and a production restriction using the framework of Weitzman (1974). We model a production restriction as an increase in the slope of the supply, because a policy like the one in Section 2 removes production capacity, thus limiting supply’s response to prices. Unlike a quota, the policy does not fix output. We show that, as a result, uncertainties about demand, supply and externality can all matter for the welfare ranking of policy instruments.

3.1 Setup

Social welfare from a regulated activity with output q is $W = B(q) - C(q)$, where B is social benefit (private surplus minus external damages), C is private cost. The optimal quantity is $\hat{q}$ such that $B' = C'$. To analyze the welfare effects of policy instruments that shift quantities near $\hat{q}$, we follow Weitzman (1974) and take a quadratic approximation approach. Let $x \equiv q - \hat{q}$ denote the deviation from $\hat{q}$. The welfare difference is approximately

$$\Delta W \equiv W(q) - W(\hat{q}) \approx (\beta - \alpha)x - \frac{D}{2}x^2, \quad (1)$$

where α and β are mean-zero, independent shocks to marginal cost and marginal benefit at $\hat{q}$ with variances σ_α^2 and σ_β^2 , and $D \equiv C'' - B''$. We assume that the slopes of marginal benefit and cost $B''(q) < 0, C''(q) > 0$, and thus $D > 0$. In an application where quantity generates negative externality, β captures shocks to marginal private benefit in demand and marginal damage from externality.

3.2 Policy Instruments

A policy maker chooses an instrument before shocks realize to maximize expected welfare $E[\Delta W]$. The instruments differ in how quantity adjusts after shocks.

Tax The regulator announces a per-unit price t for the activity. Firms observe the realized cost and produce where marginal cost equals this price: $C_1(q, \theta) = t$. Maximizing $E[\Delta W]$

¹⁵We plot the total output in Appendix C.1.

over t yields $t = E[C'(\hat{q})] = E[B'(\hat{q})]$. Given the realized shock α , the quantity adjusts so that the marginal cost matches t , and the realized deviation of quantity from $\hat{q}$ is:

$$x_{\text{tax}} = -\alpha/C'' . \quad (2)$$

Quota The regulator fixes output at $\bar{q}$. Maximizing $E[\Delta W]$ over $\bar{q}$ yields $\bar{q} = \hat{q}$: $x_{\text{quota}} = 0$.

Production Restriction For a given restriction that raises the marginal-cost slope from C'' to $C'' + \kappa$ ($\kappa > 0$), quantity adjusts to equate the marginal benefit to the steeper marginal cost after shocks:

$$x_{\text{restriction}} = \frac{\beta - \alpha}{D + \kappa} . \quad (3)$$

As $\kappa \rightarrow \infty$, $x_R \rightarrow 0$ and the restriction converges to a quota.

3.3 Welfare Ranking

Substituting each instrument's x into (1) and taking expectations yields the following comparisons.

Tax vs. Quota (Weitzman, 1974)

$$E[W_{\text{tax}} - W_{\text{quota}}] = \frac{C'' + B''}{2(C'')^2} \sigma_{\alpha}^2 . \quad (4)$$

A tax dominates when marginal cost is steep relative to marginal benefit ($C'' + B'' > 0$); a quota dominates when marginal benefit is sufficiently steep ($C'' + B'' < 0$). Only cost uncertainty σ_{α}^2 enters because the quota eliminates all quantity variation.

Tax vs. Production Restriction

$$E[W_{\text{tax}} - W_{\text{restriction}}] = \underbrace{\frac{C'' + B''}{2(C'')^2} \sigma_{\alpha}^2}_{\text{tax vs. quota}} - \underbrace{\frac{D + 2\kappa}{2(D + \kappa)^2} (\sigma_{\alpha}^2 + \sigma_{\beta}^2)}_{\text{restriction vs. quota}} . \quad (5)$$

Equation (5) extends (4) in two ways. First, benefit-side uncertainty σ_{β}^2 now enters: because the production restriction does not fix quantity, output responds to demand and damage shocks, and their variance affects welfare. Second, the comparison decomposes into the Weitzman advantage of a tax over a quota ($E[W_{\text{tax}} - W_{\text{quota}}]$) minus the advantage of the restriction over a quota ($E[W_{\text{restriction}} - W_{\text{quota}}]$). The second term is always positive because,

locally, the restriction dominates a pure quota for allowing partial quantity adjustment to shocks. As $\kappa \rightarrow \infty$, the second term vanishes and (5) reduces to (4).

3.4 Welfare Weights

We now consider a policy objective function that weighs components of the welfare function differently. We specialize the analysis to the empirical context in Section 2, where the government prioritizes improving producer surplus and reducing pollution (external damage). We focus on comparing tax and production restriction.

Under a tax $t > 0$, the government with weight w_{PS} on producer surplus and w_L on external damages evaluates the tax at

$$\Omega_{\text{tax}} = CS + w_{PS}(R - C - tq) + tq - w_L L, \quad (6)$$

where CS , R , tq and L are consumer surplus, revenue, tax revenue and the external damage. A production restriction generates no tax revenue:

$$\Omega_{\text{restriction}} = CS + w_{PS}(Pq - C) - w_L L. \quad (7)$$

We now sign the difference between the objective values under the tax and production restriction for higher w_{PS} or w_L .

Producer weight. The envelope theorem implies that

$$\frac{\partial E[\Omega_{\text{tax}} - \Omega_{\text{restriction}}]}{\partial w_{PS}} = E[(Pq - C - tq)_{\text{tax}} - (Pq - C)_{\text{restriction}}] \approx -t\hat{q} < 0. \quad (8)$$

The derivative shows that production restriction is more likely to be the better instrument when the government values producer surplus. The intuition is that the tax revenue $t\hat{q}$ extracted from producer surplus (weighted by $w_{PS} > 1$) is devalued when converted to government revenue (weighted by one).

Damage weight. Decompose $\beta = \beta_d - \beta_\ell$, where β_d and β_ℓ are independent shocks to marginal private benefit and marginal damage, with $\sigma_\beta^2 = \sigma_d^2 + \sigma_\ell^2$. Then

$$\frac{\partial E[\Omega_{\text{tax}} - \Omega_{\text{restriction}}]}{\partial w_L} = \underbrace{-\frac{\sigma_\ell^2}{D + \kappa}}_{\text{damage-shock response}} - \frac{L''}{2} \left(\frac{\sigma_\alpha^2}{(C'')^2} - \frac{\sigma_\alpha^2 + \sigma_\beta^2}{(D + \kappa)^2} \right). \quad (9)$$

The first term shows that the restriction (Eq. (3)) has a greater first-order effect on output compared with a tax, lowering expected damage through less output. The second term may mitigate this effect when the tax generates less output variance as the restriction $(\sigma_\alpha^2/(C'')^2 < (\sigma_\alpha^2 + \sigma_\beta^2)/(D + \kappa)^2)$, which limits the chance of catastrophic damage under tax when the loss function is sufficiently convex.

Together, a greater weight on producer surplus unambiguously rationalizes the production restriction, while whether a greater weight on external damages does so depends on the relative magnitudes of cost and benefit uncertainty. Our empirical exercise estimates demand and supply for coal in China and asks which combinations of welfare weights can explain the observed policy choice.

4 Data

4.1 Data Sources

Our primary dataset comes from the China Coal Transportation and Distribution Association (CCTDA), a major trade group in the coal industry. The data include monthly shipment quantities of coal between provinces from 2012 to 2016. The tonnage data combine raw coal, washed coal and other derivative products with high coal content (such as coke and briquette).¹⁶ The shipment data include shipments within a province and to other provinces.

We then compile additional data from other sources. We collect province-level mining capacity data from 2014 to 2018 from the National Energy Administration and National Mine Safety Administration. Any capacity change is approved every 6 months.¹⁷ We also obtain province-month production data in 2014–17 from CCTDA and 2018–2019 from National Bureau of Statistics (NBS). The production data reflect the total tonnage of excavated raw coal. We obtain monthly production data for key state-owned (SOE) coal mines from CCTDA, which are the largest coal mines in each province.¹⁸ The monthly coal imports, measured in total tonnage, from outside China to each province between 2014 and 2017 are

¹⁶Consistent with the standard NBS practice, the data account for transportation of the tonnage of final end products from the raw coal producers to the end users. More specifically, if 1 ton of raw coal was extracted in province *A*, transported to province *B* for washing (removing impurities) with quantity reduced to $x < 1$, and burned at power plants in province *C*, the data would only include x in transportation from *A* to *C* and not count the *A* – *B* transportation. We also note that not all raw coal needs washing, and many coal mines are integrated with washing facilities and need not transport coal to a third province for processing. Our data closely match the NBS aggregate production and consumption. See Appendix B for more details.

¹⁷In principle, the annual production should not exceed the approved level, and monthly production should not exceed 10% of the 1/12 of the annual level.

¹⁸NDRC designates an SOE coal mining firm as a “key SOE” based on a number of criteria, which include capacity, reserves, mining technology and coal quality (whether the coal is mainly for electricity production).

from the firm International Coal.¹⁹ We collect monthly provincial coal prices in 2014–2019 from NDRC.²⁰ To study the impact of coal consumption on air pollution, we collect air quality data (hourly concentration measures of pollutants such as PM2.5 and PM10) at the monitoring station level from the Ministry of Environmental Protection.

Importantly, we do not directly have data on monthly province-level coal consumption. To construct coal consumption at the province-month level, we impute the quantities based on (1) shipment data from CCTDA, (2) monthly output of major industries using coal, and (3) annual province-level coal consumption, both from NBS. Details of this imputation are in Appendix C.

To verify the consistency of data from different sources, we compare the CCTDA’s shipment data out of a province with the production data from NBS in Appendix B. Our shipment data are slightly lower (by about 4% to 8%) than the NBS statistics. At a more granular level, the same pattern holds consistently across provinces.

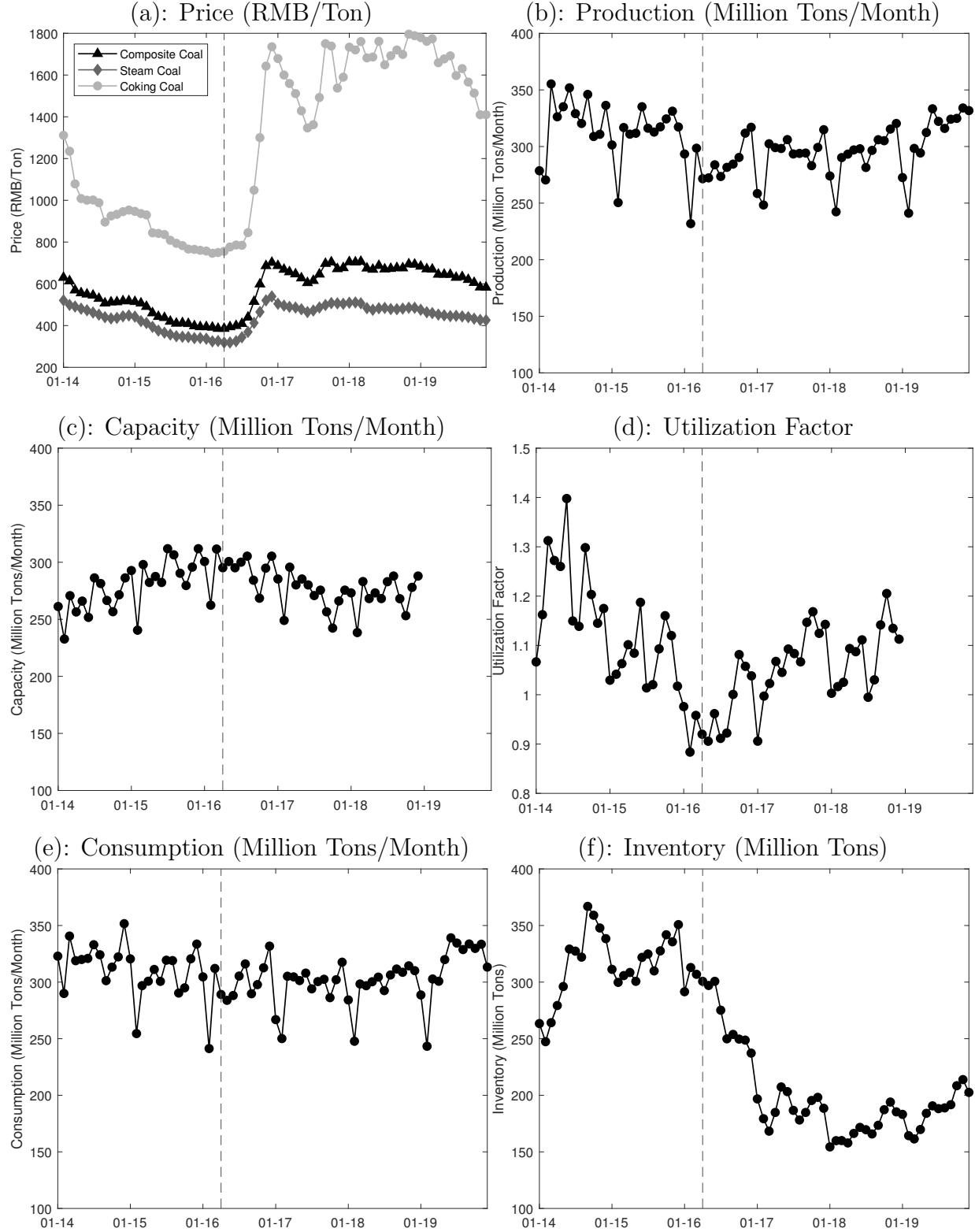
Data Aggregation

In the rest of the paper, we use the total weights of coal produced and consumed to analyze supply and demand. We do not further differentiate among coal types within the quantity data. Given our research questions, several potential concerns about the data may warrant discussion. Importantly, to what extent can a single demand function capture the composite demand for a variety of coal and coal products combined in our quantity data? What is the appropriate price for this composite demand? We note that in China, the demand for different coal products is likely highly correlated. The widespread use of coal blending techniques enables power plants to substitute across different coal types in the event of shortages. Other end users such as heating facilities are likely even less sensitive to coal quality. As we will show next, thermal and coking coal prices are highly correlated. Therefore, we construct an average price for each province and month based on the two price series weighted by their consumption shares.

¹⁹The firm provides the data through the website <https://mcoal.in-en.com/>. We do not consider China’s exports, which account for less than 0.2% of coal production during our sample period.

²⁰The publicly available coal price data are based on coal used for electricity generation and heating. This type of coal, which includes a variety of coals with different heat contents and purity, is collectively referred to as “steam coal” or “thermal coal.” We collect monthly province-specific prices of steam coal using the China Steam Coal Price Index from the Price Monitoring Center of NDRC. The index is based on data collected from 1,600 firms that rely on coal as the main fuel source, including major coal-fired power plants and heating facilities. Prices of coal used for coking are based on prices in Ganqimaodu in Inner Mongolia, a major hub of coking coal distribution.

Figure 1: Time-Series Data of the Coal Market



Notes: Panel (a) plots the prices of steam coal, coking coal and the composite price, (b) plots the national monthly production, (c) plots the monthly national capacity, which is adjusted from the capacity that changes every 6 months for the number of working days in each month (our capacity data end at the end of 2018), (d) is the utilization factor, defined as national monthly production divided by capacity, (e) is the national monthly consumption, and (f) is the national inventory. The vertical line indicates the start of the supply reduction policy.

4.2 Data Summary

4.2.1 National Time Series

We plot the time series of the national coal prices, production, capacity, utilization factor, inventory and consumption in Figure 1. Coal prices shown in (a) increased significantly for both coking and steam coal as the policy started in Shanxi in April 2016. The prices remained higher throughout 2017 and 2018 and gradually decreased in 2019 after the policy pressure eased. We plot the monthly production in (b). The production in 2016 was noticeably lower than similar periods both before and after the policy.

The supply squeeze was a result of two aspects of the policy. First, the government closed coal mines and suspended new approvals, resulting in a net decrease in capacity as shown in (c). Second, the campaign to reduce over-production above approved capacity levels directly limits output. In (d), we plot the utilization factor, defined as the production-to-capacity ratio. This ratio in 2017 and 2018 (post-policy) is lower than or comparable to 2014–15, despite significantly higher prices in 2017 and 2018. The utilization factor is also lower in 2016 than in 2017–2018, reflecting more stringent enforcement in 2016 (the 276-working-day policy).

In (e) and (f), we find that the market went through a transition in 2016. The figure (e) shows that consumption has a more muted response in 2016 than production. Excess demand over production corresponds with an over 40% decrease in inventory (Figure 1.(f)) during the rapid price increase in 2016.²¹ The inventory stabilized at a lower level in 2017 and 2018, when production increased above 300 million tons per month, matching the level of consumption.

²¹Monthly foreign imports also increased by about 10 million tons per month in 2016 and remained at about 20 million tons per month post-policy.

Table 1: Summary Statistics

	Mean	SD	p25	Median	p75
<i>Panel A. Coal-Consuming Provinces</i>					
Consumption (Million Tons/Month/Province)	10.16	7.63	4.62	7.47	13.30
Price (RMB/Ton)	532.01	133.87	430.21	523.03	615.45
Wind Generation (TWh/Month/Province)	0.53	0.73	0.06	0.29	0.70
Cooling Degree Days ($^{\circ}\text{C}$ /Month/Province)	17.77	38.16	0	0	11.12
N=1440; based on 30 coal-consuming provinces in 2014–2017					
<i>Panel B. Coal-Producing Provinces</i>					
Quantity (Million Tons/Month/Province)					
Supply	11.30	19.91	1.20	3.61	9.98
Capacity	11.21	17.85	1.59	5.18	12.40
N=1200; based on 25 coal-producing provinces in 2014–2017					
<i>Panel C. Foreign Import</i>					
Quantity (Million Tons/Month)	18.12	4.97	14.00	17.55	21.82
N=48; 2014–2017					

Notes: Panel A shows summary statistics for 30 coal-consuming provinces. The demand consists of shipments from Chinese provinces and foreign imports. Coal consumption is aggregated from consumption in all sectors using coal for each province (Appendix C). The price is the quantity-weighted average of thermal and coking coal prices plus a calibrated transportation cost (Appendix E) based on coal origins. We also use monthly wind generation and cooling degree days in the analysis. Panel B reports summary statistics for 25 coal-producing provinces. Supply is the sum of all shipments from that province each month. Capacity is aggregated from the mine-level data from NEA. Panel C reports total imports.

4.2.2 Summary Statistics

For the main analysis of the paper, we focus on the years from 2014 to 2017.²² Table 1 summarizes our main data set. There are 30 coal-consuming provinces and 25 coal-producing provinces. All coal-producing provinces also consume coal. The reported prices include transportation costs based on coal origins, which account for approximately 17.5% of the composite price.²³

²²CCTDA’s cross-province shipment data are available through 2016, and its production data are available through 2017. We cross-validate both datasets using NBS statistics. For subsequent years, NBS reports province-level monthly production, except for January and February.

²³Appendix E details how we calibrate the transportation costs.

Table 2: Mean and Standard Deviation of Cross-province Shipment Shares, 2014–2016

	Inner Mongolia	Shanxi	Shandong	Hebei	Jiangsu	Shaanxi	Henan	Liaoning	Guizhou	Xinjiang
Inner Mongolia	0.392 [0.014]	0.011 [0.002]	0.049 [0.008]	0.017 [0.003]	0.064 [0.010]	- -	0.002 [<0.001]	0.082 [0.003]	- -	- -
Shanxi	0.006 [0.002]	0.298 [0.013]	0.104 [0.007]	0.231 [0.029]	0.103 [0.012]	0.002 [0.001]	0.055 [0.012]	0.017 [0.002]	- -	0.001 [0.001]
Shaanxi	0.004 [0.001]	0.148 [0.020]	0.086 [0.049]	0.039 [0.010]	0.101 [0.027]	0.411 [0.070]	0.059 [0.007]	- -	0.001 [0.001]	- -
Guizhou	-	-	-	-	-	-	-	-	0.740 [0.025]	-
Shandong	-	-	0.857 [0.016]	0.012 [0.005]	0.033 [<0.001]	-	0.008 [0.001]	0.002 [<0.001]	- -	- -

Notes: The table reports the shipment shares of the five largest coal-producing provinces to the ten largest coal-consuming provinces, averaged over 36 months between 2014 and 2016. The share is calculated as a month's shipment from i to j divided by the total shipment from i in the month. The standard deviations are reported in square brackets.

4.2.3 Cross-province Shipment Shares Are Stable Over Time

We show that the share of a province’s coal production transported to another province is highly stable over time. This important empirical fact will guide the modeling decisions in the model. In Table 2, we report the mean and standard deviation (in square brackets) of the shipment shares from province i to province j , defined as $\frac{Q_{ijt}}{Q_{it}}$, where Q_{ijt} is the tonnage of coal produced in i and transported to j , and Q_{it} is the total production in i in month t . We list the five largest coal-producing provinces on the row and the ten largest coal-consuming provinces across columns. The standard deviation for the majority of province pairs is negligible relative to the mean, indicating a high degree of stability in shipment shares over time.

There are two possible reasons for this stability despite substantial variation in coal prices over time. First, the government and railway operators allocate railway capacity when coal buyers and sellers enter into long-term contracts. The rail capacity allocations create rigidity within the duration of long-term contracts, as any significant rerouting of shipment across provinces could create cascading disruptions in the interconnected rail network. This constraint incentivizes coal buyers to switch between suppliers within a given province even when coal buyers procure coal from outside their long-term contracts. Second, although power plants can substitute across different grades of coal to some extent through coal-blending technology, they still tend to prefer similar coals when purchasing on the spot market, because coal blending can add to the plants’ operating costs (Zhao, 2021). Since mines within the same province are more likely to produce similar grades of coal, they may supply to the same set of power plants, consistent with stability across multiple years.

We also find support for these structural reasons in explaining the rigidity of shipment shares. Between 2014 and 2016, the log difference between the highest and lowest coal prices across provinces varies between 0.62 and 0.94.²⁴ If shipment shares could change flexibly in response to price differentials, we would expect to observe strategic shipping behaviors to exploit the arbitrage opportunities, keeping price dispersion relatively constant. Instead, the observed persistence of price differentials points to a lack of such strategic behaviors. Our model below thus focuses on production decisions as opposed to shipment destination choices.

²⁴The log difference of 25th–75th price levels ranges from 0.12 to 0.35.

5 Coal Demand

We specify coal consumption in province j and month t as

$$\ln q_{jt}(p_{jt}) = \alpha^D \ln p_{jt} + \beta_X X_{jt} + F E_j^{\text{demand}} + F E_{m(t)}^{\text{demand}} + \varepsilon_{jt} \quad (10)$$

where the quantity q_{jt} corresponds to the consumption of coal in province j , and p_{jt} represents the composite coal price in the province.²⁵ The parameter α^D has the straightforward interpretation of price elasticity. The covariates include province j 's one-month lagged cooling degree days (CDD) and wind generation, which affect the local electricity residual demand for coal power plants, and an indicator for 2017 to capture government intervention to shore up long-term contract compliance on the demand side. We also include fixed effects for the province and the month of the year. The term ε_{jt} captures the unobserved demand shock.

We treat wind generation as largely exogenous to the unobserved shocks ε_{jt} for two reasons. First, wind power build-out is primarily driven by centrally planned investment that targets areas rich in wind resources (State Council, 2014). Second, NDRC directs local governments and the power grid to guarantee priority dispatch and full purchase of wind generation (NDRC, 2016b). Given these policies, we assume wind production to be driven by an exogenous long-term trend plus local and short-run weather conditions, which generates exogenous variations in residual demand for coal generation.

Static Demand Assumption. In theory, the coal purchase decisions can be dynamic, where coal users can use existing inventory to smooth consumption when prices change (Jha, 2023). Our demand function is static, because coal consumers do not store large quantities of coal in our context. Specifically, NDRC requires coal power plants to keep enough stock for at least 7 days of operation, but explicitly caps inventories at no more than 12 days in peak seasons (NDRC, 2021a). A historically high value in 2024 is just 27 days (CCTV News, 2024). Major steel plants keep a similar level of coal inventory (BOCI, 2017). This practice contrasts sharply with the norm in the US, where over 90% of coal power plants keep enough coal to generate electricity for over 60 days (EIA, 2023). We also note that, although there may still be arbitrageurs with sizable capacity, such as dedicated coal traders storing coal at ports, the model in (10) focuses on coal consumption by end users instead of shipment. In our counterfactual analysis, we focus on the “steady state” of the market, where demand matches the level of consumption in the markets before 2016 or in 2017, and the inventory

²⁵We note that there may exist potential differences between spot prices and the re-adjusted prices facing buyers and sellers on long-term contracts. Because we do not have contract prices or order-level prices, we use the spot prices as a first-order approximation for sales in a province.

level was roughly stable over time (Figures 1.(e) and (f)).

Long-term Contracts. Another concern is that long-term contracts may reduce demand for spot-market coal. Section 2 shows that this is largely not the case for 2014-2016 when delivery prices and quantities significantly deviated from contracts: buyers frequently purchased from the spot market when market prices were low, and sellers sold to the market demand when prices were high. In 2017, government measures both improved contract compliance while also bringing the contract prices more in line with spot prices, reducing the distinction between demand for contracted and spot market coal. In addition to including a 2017 indicator in demand, we further explore the robustness of our demand estimates in Appendix G where we assume that contracts proportionally reduce the demand for spot market coal.

5.1 Identification

The equilibrium price may be correlated with the unobservable ε_{jt} . We construct shift-share instrumental variables that exploit the timing and exposure differences of provinces to the production restrictions as documented in Section 2.3.1.

Instrumental Variables (IV)

There are three potentially useful variations:

1. The policy started first in Shanxi in April 2016, followed by other provinces in May. Shanxi is also one of the provinces that experienced the largest decrease in output.
2. The policy imposes stricter monitoring programs on the production of state-owned enterprises (SOEs), and the share of production by SOEs differs across provinces.
3. The policy closed small coal mines, and the cutoff thresholds differ across provinces.

In each province, the variations in policy timing and intensity shift the local supply through the largely fixed shipment shares (Section 4.2.3). Importantly, because the demand function (10) already accounts for policy effects through the 2017 indicator, our excluded instruments are based on purely supply-side variations in 2016, when the policy rolls out across provinces at different times and enforcement intensity differs from 2017 (the 276-working-day policy in 2016). Specifically, we use χ_{it} to denote whether the policy has applied to province i 's supply in month t , and for each province j in month t , we define (1) a timing IV

$$Z_{jt}^{\text{Timing}} = \sum_i w_{ij}^{2012-2013} \chi_{it}, \quad (11)$$

where $w_{ij}^{2012-2013}$ is the coal shipped from i to j divided by the total shipment to j , averaged across months from 2012 and 2013 (before the estimation sample), which captures j 's historical dependence on coal from i .²⁶ We also define (2) an SOE IV,

$$Z_{jt}^{\text{SOE}} = \sum_i o_i w_{ij}^{2012-2013} \chi_{it}, \quad (12)$$

where o_i is the output share²⁷ of key SOEs in province i , also averaged across months from 2012 and 2013. In the above, the interactions between o_i and $w_{ij}^{2012-2013}$ proxy the exposure differences to the supply policy shocks. Finally, we define (3) a capacity IV,

$$Z_{jt}^{\text{Capacity}} = \sum_i K_i^{\text{Small}} w_{ij}^{2012-2013} \chi_{it}, \quad (13)$$

and K_i^{Small} is province i 's capacity in January 2014 below the province's cutoff thresholds for closure during the policy in 2016.²⁸

IV Validity

Our IVs exploit the independence between demand shocks and the shipment shares' interaction with χ_{it} conditional on province fixed effects and month-of-the-year fixed effects, which is the key assumption for the validity of Bartik-type shift-share instruments (Goldsmith-Pinkham, Sorkin and Swift, 2020). The assumption relies on the fact that the variations in policy timing (the policy started in Shanxi first) and levels of reductions (SOEs' production is more restricted) are primarily due to policymakers targeting the largest coal mines to produce fast results (Appendix A), as opposed to correlations with demand shocks. In particular, the two largest provinces' SOEs significantly reduced their production in 2016 (Appendix I.2).

One potential concern with the exclusion restriction of the shift-share IVs is a correlation between policy start times and other economic policies. We note that the 2015-2016 economic stimulus plan is a nationwide policy that generally eases the borrowing constraints for transportation projects, such as the national network of high-speed rails (HSR). A corre-

²⁶We find that the shares of shipment from a province, w_{ij} , just like the shares of shipment to a province in Section 4.2.3, are stable over time during our sample. We argue in Section 4.2.3 that structural reasons not due to changing prices are likely responsible for the rigidity.

²⁷Ideally, we should use capacity shares to construct the IV because there can be potential serial correlations in unobservables. We do not have capacity data for SOE mines. The main assumption is that any serial correlation of production between 2013 and 2016 is negligible.

²⁸Following the official policy document, the capacity cutoffs for "small" coal mines to be closed are 0.3 million tons/year for major coal-producing provinces: Shanxi, Inner Mongolia, Shaanxi, and Ningxia, 0.15 million tons/year for Hebei, Liaoning, Jilin, Heilongjiang, Jiangsu, Anhui, Shandong, Henan, Gansu, Qinghai, and Xinjiang, and 0.09 million tons/year for the remaining provinces.

lation with our instruments may arise when the first province to enact the policy, Shanxi, is also a major supplier to provinces with more infrastructure projects in April that would use more coal. Appendix D lists major active infrastructure construction projects in April and May of 2016. The large volumes of concrete and steel used in these projects could generate substantial demand for coal and coke. We find no clear correlation between the locations of these projects and Shanxi’s main destination provinces—Shanxi, Hebei, Shandong, and Jiangsu—which together account for over 70% of Shanxi’s output (Table 2).

Another concern is that policy timing may be endogenous if coal buyers anticipated the policy and stockpiled coal in advance. Policy guidance and common industry practices discussed previously show that major end users of coal do not have significant storage capacity. National coal inventory data from NBS in Figure 1 show that the inventory actually declined before the policy, suggesting that the policy is likely a surprise.

5.2 Estimation

To understand the strengths of the IVs and the underlying variations they capture, we report the first-stage results in Table 3. Columns (1)–(3) regress the natural log of prices on each instrument in the previous section. We find that the results are intuitive. Shanxi’s early policy implementation raises the prices in provinces more exposed to its supply. Similarly, prices are higher in provinces where a greater mix of supply historically comes from SOEs or small mines, and given the timing difference, particularly Shanxi’s SOEs and small mines. The F statistics indicate that the timing and SOE IVs are considerably stronger.

Table 4 reports the second-stage results based on the instruments in Table 3. We use the estimated elasticity of -0.635 based on the timing and SOE IVs in column (5) as our primary specification. Notably, the stronger IVs in columns (2), (3) and (5), based on policy timing, SOEs and their combination, all imply similar demand elasticities. Although the estimate based on the weaker IV of small mine capacity (column (4)) is larger, the estimate in our primary specification is not rejected at the 95% confidence level. The IV estimates are considerably different from the positive OLS estimate in column (1). In comparison, prior work (Burke and Liao, 2015) has estimated the elasticity for coal demand to be between -0.3 and -0.7. Finally, we note that our results are robust to using the different IV estimates.

Table 3: Demand Estimation: First-Stage

	(1) Timing	(2) SOE	(3) Small-Mine Capacity	(4) Timing+SOE
Z_{jt}^{Timing}	0.075*** (0.014)			0.017 (0.029)
Z_{jt}^{SOE}		0.147*** (0.023)		0.117** (0.049)
Z_{jt}^{Capacity}			0.086** (0.034)	
Wind Generation (TWh)	-0.067*** (0.025)	-0.066*** (0.025)	-0.056*** (0.021)	-0.067*** (0.025)
CDD/1000	0.260*** (0.084)	0.249*** (0.085)	0.335*** (0.100)	0.249*** (0.085)
Province FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
Kleibergen-Paap F Statistic	25.46	37.84	6.057	19.65
N	1440	1440	1440	1440

Notes: The table reports the first-stage results based on various IVs. The dependent variable is the ln price in each province and month.

	OLS	Second-Stage Results Based on Each Set of IV's			
	(1)	(2) Timing	(3) SOE	(4) Small-Mine Capacity	(5) Timing+SOE
$\ln p_{jt}$	0.115*** (0.031)	-0.696*** (0.218)	-0.619*** (0.201)	-1.258*** (0.478)	-0.635*** (0.203)
Wind Generation (TWh)	-0.039** (0.016)	-0.081*** (0.026)	-0.077*** (0.023)	-0.110*** (0.039)	-0.078*** (0.024)
CDD/1000	0.465*** (0.219)	0.742*** (0.237)	0.716*** (0.229)	0.934*** (0.296)	0.722*** (0.231)
Province FE	YES	YES	YES	YES	YES
Month FE	YES	YES	YES	YES	YES
N	1440	1440	1440	1440	1440

Notes: The table reports OLS and IV estimates of the demand model. The dependent variable is the log of demand (shipment into a province and imports). Column (1) is the OLS result. Columns (2)-(5) are results based on various IVs.

6 Coal Supply

Our supply model incorporates two key institutional features of China's coal industry. First, the coal market is not concentrated. In Appendix I.3, we plot the combined output shares of the three largest coal-mining firms (based on the output in the key SOE production data) in each of the top five coal-producing provinces. No single firm has more than 30% of the output share in a province, and the share of the third largest firm is often below 11%. Second, as we show in Section 5.1, the share of supply from one province to another is highly stable over time.

Motivated by these two observations, we specify the aggregate marginal cost function at the province level as

$$\ln mc_{it} = \alpha^S \frac{q_{it}}{K_{it}} + FE_i^{\text{supply}} + FE_{m(t)}^{\text{supply}} + \xi_{it}, \quad (14)$$

where K_{it} is the capacity in province i in month t , which is taken as given in the data, and we include fixed effects at the province and month level. We assume that, in equilibrium, the marginal cost matches the effective price $P_{it} = \sum_j s_{ij} (p_{jt} - \tau_{ij})$, where s_{ij} is the share of coal shipped from i to j in the total shipment from i ,²⁹ p_{jt} is the price in the destination province j , and τ_{ij} is the transportation cost calibrated following the procedure in Appendix E.³⁰

Behavioral Foundation

The marginal cost function (14) implies the supply curve $\frac{Q_{it}(P_{it})}{K_{it}} = \gamma \ln P_{it} + \widetilde{FE}_i^{\text{supply}} + \widetilde{FE}_{m(t)}^{\text{supply}} + \tilde{\xi}_{it}$, where $\gamma = \frac{1}{\alpha^S}$, with fixed effects and shocks also appropriately transformed. The supply function can be microfounded as the aggregate decisions of many atomistic, competitive mines. Suppose each producer in province i sells a share s_{ij} to province j . The cost function is given by $c(q; \theta)$, where the cost parameter θ differs across mines and follows the distribution Θ_i . The total mass of the mines is $\mathcal{M}$. Slightly abusing notation, we use $a_{\theta t}$ to denote this producer's output decision, which satisfies the first-order condition $P_{it} = c'(a_{\theta t}; \theta)$. Under the assumptions that the marginal cost function c' is strictly monotonic in $a_{\theta t}$ for any θ , we can then invert the production decision as $a_{\theta t} = c'^{-1}(P_{it}; \theta)$. Integrating both sides with respect to $\Theta_i(\theta)$ and multiplying the total mass of the mines $\mathcal{M}$ yield an aggregate supply curve as a function of P_{it} , and appropriate choices of c and Θ_i would give the specific exponential form.

²⁹We use the average shares as in Table 2, and the results are similar if we use just any one year's data to construct the shares.

³⁰We also include the resource tax in the effective price.

A caveat of this microfoundation is that the effective price P_{it} would no longer be a sufficient statistic for all relevant prices if the output share s_{ij} is heterogeneous across individual mines. In the extreme case, each mine may supply coal to only one destination province, as opposed to splitting its output across provinces. Although we do not have mine-level data on output and shipment, we can rule out this extreme case based on news reports where large mines routinely supply to buyers in multiple provinces (Xinhua, 2016; China Securities Journal, 2017; Xinhua, 2018).

Static Supply Assumption

We assume that the coal output decisions are static. According to our interviews with industry practitioners, mine owners plan production based on current prices, and dynamic factors such as the amount of unexcavated coal or interest rates do not appear to be first-order drivers of their production decisions. For context, the coal reserves in China are close to 150 billion tons, while the annual production varies between 3 and 4 billion tons, with a similar amount of new reserves discovered every year.

Supply Function After Policy

Given that we observe only 25 provinces and the enforcement intensities differ in 2016 and 2017, our model of the policy effects is necessarily parametric. Our baseline specification allows the policy to raise marginal costs more sharply once production approaches approved capacity:

$$\ln mc_{it} = \alpha^S \frac{q_{it}}{K_{it}} + FE_i^{2016 \text{ policy}} + \alpha^{S, \text{policy}} \chi_{it}^{2017} \left(\frac{q_{it}}{K_{it}} - \nu \right)_+ + FE_i^{\text{supply}} + FE_{m(t)}^{\text{supply}} + \xi_{it}, \quad (15)$$

where χ_{it}^{2017} is an indicator for whether t is in 2017, $(x)_+ = \max(x, 0)$, and ν is the utilization cutoff at which the post-policy slope steepens. In the case where the post-threshold slope $\alpha^{S, \text{policy}}$ is large, the policy mimics an efficiently implemented quota within a province, and the quota level is νK_{it} . The slope is part of the parameters we estimate from data.

We note that the policy may raise the marginal costs significantly more when production is close to and past approved capacity. At the same time, the policy shifts from broad output cuts in 2016 to more targeted enforcement of overproduction in 2017 (NDRC, 2017b). Smaller and higher-cost mines are subject to more intense scrutiny, which may translate into a larger increase in aggregate marginal cost near capacity. We therefore also estimate an alternative specification in which Shanxi and Inner Mongolia are allowed to have a different

post-threshold slope from the remaining provinces:

$$\begin{aligned} \ln mc_{it} = & \alpha^S \frac{q_{it}}{K_{it}} + FE_i^{2016 \text{ policy}} + \alpha^{S, \text{policy}} \chi_{it}^{2017} \left(\frac{q_{it}}{K_{it}} - \nu \right)_+ \\ & + \alpha^{S, \text{policy, top 2}} \chi_i^{\text{Top 2}} \chi_{it}^{2017} \left(\frac{q_{it}}{K_{it}} - \nu \right)_+ + FE_i^{\text{supply}} + FE_{m(t)}^{\text{supply}} + \xi_{it}, \end{aligned} \quad (16)$$

where $\chi_i^{\text{Top 2}}$ is an indicator for whether province i is Shanxi or Inner Mongolia.

6.1 Identification

The supply function and the resulting estimation equation can be written in the general form

$$\ln P_{it} = g(q_{it}, X_{it}^{\text{supply}}) + \xi_{it}, \quad (17)$$

where q_{it} is correlated with ξ_{it} , and there is a vector of exogenous covariates X_{it}^{supply} that accounts for province capacities, locations and time. A sizable literature (e.g., Newey and Powell, 2003) studies the nonparametric identification and estimation of models in this form. In addition to requiring instruments that are mean or quantile independent of the unobservables, the instruments should induce sufficient variations in q . We use demand shifters in (10) as instruments for the supply (Wright, 1928). The excluded instruments include wind generation, the one-month lagged CDD and the exponential of demand residuals, which we assume to be orthogonal to shocks to coal's production costs (MacKay and Miller, 2024; Döpper et al., 2024). We additionally include their interactions with a post-policy indicator. To estimate the more flexible model in (16), we further include these instruments' interactions with top 2 province indicators.

Table 5: Supply Estimation

	(1) Baseline	(2) Top 2 and Rest
α^S	1.363*** (0.525)	1.179*** (0.223)
$\alpha^{S,\text{policy}}$	1.146** (0.513)	0.666** (0.305)
$\alpha^{S,\text{policy, top 2}}$		1.439*** (0.360)
ν	0.461*** (0.104)	0.461*** (0.070)
Province FE	YES	YES
Month FE	YES	YES
N	1200	1200

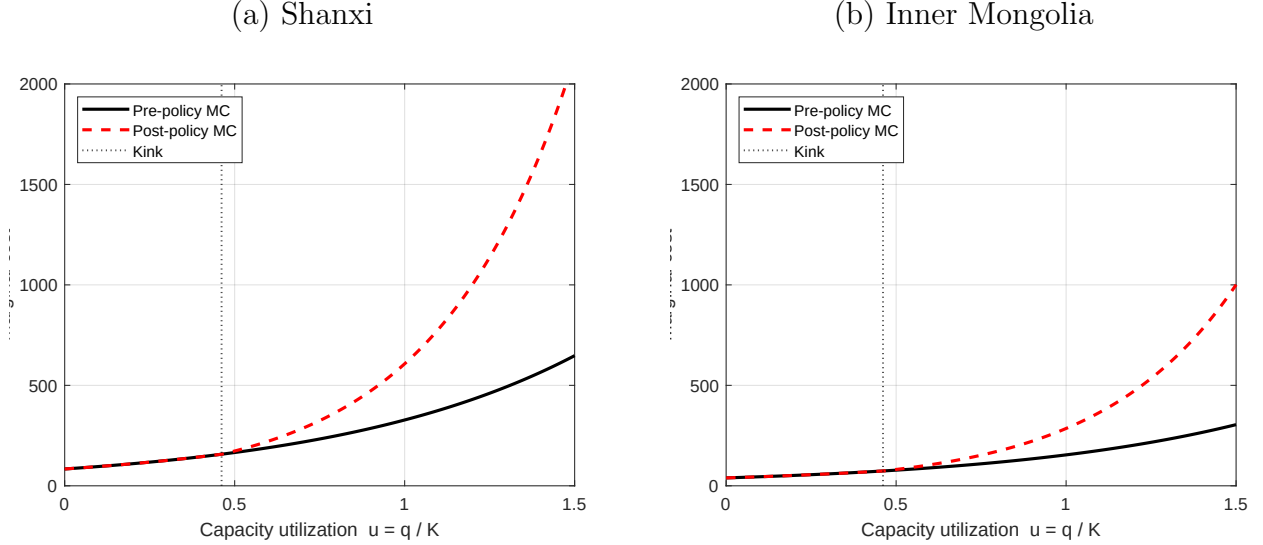
Notes: The table reports GMM estimates of the supply model. Prices faced by domestic producers in each province are constructed as weighted averages of destination prices net of calibrated transportation costs, where the weights correspond to each destination’s share in total shipments from the province. Column (1) is the baseline threshold model. Column (2) allows the post-threshold slope effect to differ between Shanxi and Inner Mongolia and the remaining provinces. The excluded instruments are shipment-share-weighted wind generation, cooling degree days, and demand residuals, together with their interactions with the post-policy indicator; column (2) also interacts these instruments with a top-2 province indicator. Standard errors are clustered at the province level and combine the supply-side bootstrap with an additional demand-side bootstrap component that rebuilds the residual IV. *, **, and *** indicate significance at the 10, 5, and 1 percent levels.

6.2 Estimation

Column (1) of Table 5 reports the estimates of the baseline supply function in (15), and column (2) reports the more flexible model in (16). In the baseline model, the pre-policy slope is 1.363 and the post-policy increment above the utilization threshold is 1.146, with an estimated threshold of $\nu = 0.461$. In column (2), the estimated post-threshold slope increment is 0.666 for the non-top-2 provinces, with a further increment of 1.439 for Shanxi and Inner Mongolia and the same threshold estimate. To visualize the cost increases, Figure 2 plots the aggregate marginal cost functions for Shanxi and Inner Mongolia before and after the policy under the baseline model.

We interpret these increases as reflecting two factors. First, the policy may impose direct compliance costs, such as halting production for inspections. Second, by preventing low-cost mines from operating above capacity, the policy reallocates output toward higher-cost mines, shifting the production mix toward costlier producers and raising aggregate marginal costs

Figure 2: Estimated Aggregate Supply in the Two Largest Coal-Producing Provinces (Shanxi and Inner Mongolia)



The figures compare the aggregate supply functions of Shanxi and Inner Mongolia with and without policy effects in 2017.

for any given total quantity.

The estimates imply that the policy has a greater effect near capacity, especially in the largest producing provinces. This is consistent with the institutional design of the reform, which shifted from broad output cuts in 2016 to close monitoring of overproduction in 2017, and with policymakers’ emphasis on visible “breakthroughs” in key coal-producing provinces (Appendix A).

Finally, we compare the average variable costs with costs reported in financial statements of publicly listed coal mining firms in 2014–2016. We plot these costs in Appendix Figure I.6. The estimated and reported costs are well aligned, with a correlation coefficient of 0.56 for the baseline model. For example, we find that Xinjiang and Inner Mongolia have the lowest production costs, consistent with more open-pit mining in these provinces.

7 Counterfactual Simulation

We begin by defining the equilibrium given a set of shocks. Let $\mathcal{J}$ denote the set of coal-consuming provinces, $\mathcal{I}$ for the coal-producing provinces and τ_{ij} for the transportation costs.

Definition. In period t , an equilibrium consists of a set of delivery prices $\{p_{jt}\}_{j \in \mathcal{J}}$ and production quantities $\{Q_{it}\}_{i \in \mathcal{I}}$ that satisfy (10) and (17), where a province j ’s demand is given

by $q_{jt} = \sum_{i \in \mathcal{I}} s_{ij} Q_{it}$, and the price facing producers in province i is $P_{it} = \sum_j s_{ij} (p_{jt} - \tau_{ij})$.³¹

The supply reduction policy of the reform has two main effects: (1) it reduces capacity by closing small mines and halting new entry, and (2) it tightens enforcement by closely monitoring remaining mines to prevent production above approved capacity. Because we lack mine-level data on entry and exit, we take capacity as given and focus on the enforcement effects—how stricter enforcement reduces production conditional on capacity. Our estimates of economic surplus should be interpreted as the sum of consumer surplus and variable profits, which do not account for investment or fixed costs. In all simulations, we allow foreign imports to respond endogenously, following an estimated import supply function.³² Our results are based on the baseline supply model, and they are robust to using the more flexible supply model in (16).

7.1 Enforcement Effects

Our first simulation quantifies the enforcement effects of the policy. We remove the estimated shifts in supply and re-compute the expected equilibrium outcomes under the baseline threshold model. Specifically, we set the post-policy slope term in (15) to zero. We assume that the demand and supply shocks at the province level follow an AR(1) process, and we also simulate forward the processes of CDD and wind generation.³³ We report the 2.5%–97.5% range and the average of market outcomes based on 300 simulation paths.

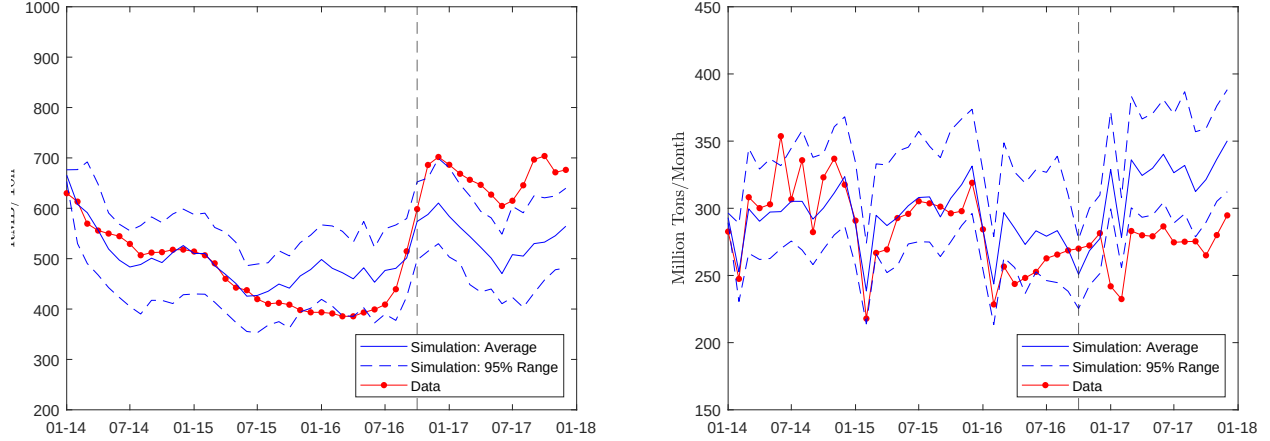
Figure 3 compares the average equilibrium prices and production under the no-policy simulation with observed outcomes under the policy, holding fixed the capacity as observed in the data. The simulations align closely with the data before the policy. We further show in Appendix I.5 that the baseline threshold model can match province-level production despite the parsimonious specification. We find that, absent the policy, prices would have been about RMB 550/ton in December 2016 as opposed to nearly RMB 700/ton in the data. The quantity would have been nearly 50% higher.

³¹We can establish the equilibrium existence by first constructing closed intervals of prices and quantities on which the equilibrium conditions map the extrema of the intervals back into the intervals using the monotonicity of the demand and supply functions and the fact that their range is $(0, \infty)$. Then an application of Brouwer’s fixed point theorem shows existence. For uniqueness, we do not find multiple equilibria in our simulations.

³²Our shipment data show that the share of imports to each province is also stable over time and allows us to construct a similar effective price for imports. We estimate the import supply elasticity in a log-log specification. An OLS regression based on data from 2014 to 2017 (48 months) yields an estimate of 1.35, with a robust standard error of 0.15. A corresponding IV estimation produces nearly identical results.

³³Appendix F reports the estimates of these processes.

Figure 3: Prices and Production With and Without the Policy
(a) Prices (b) Production



The figures plot the simulated prices and production of Shanxi and Inner Mongolia from 300 simulations in a no-policy counterfactual. We plot the simulated 2.5%–97.5% range, the average and the data paths (with policy). The vertical lines indicate the start of the policy.

Welfare Effects

Table 6 presents the welfare changes had the policy not been implemented. We focus on 2017 because inventory adjustment was unusually large in 2016: between April and December 2016, a substantial inventory drawdown contributed to the consumption by coal’s end users. By 2017, inventory levels had largely stabilized, so consumption more closely tracked production. This makes 2017 the appropriate period to apply our equilibrium framework, in which demand reflects consumption and supply maps production to consumption without a sizable inventory-change component. We simulate the market outcomes with and without the policy effects using the estimated time-series processes for shocks and other covariates conditional on their values in the last month of 2016.³⁴

Table 6 reports the no-policy outcomes relative to the observed policy. The average coal price in 2017 would be RMB 161/ton lower, domestic production would be 693 million tons per year higher, and coal consumption would be 642 million tons per year higher. Total economic surplus would increase by RMB 300 billion, with gains accruing to consumers and losses to producers. Appendix Figure I.8 breaks down the surplus changes at the province level, consistent with the policy benefiting coal-producing provinces and hurting

³⁴It is possible that the time-series processes before and after the policy may differ. We find that all of our results are qualitatively robust to using the processes estimated based on 2017 data alone.

coal-consuming provinces. At the same time, the larger quantities translate into RMB 327 billion more pollution damages from air pollution and water depletion, so overall welfare would fall by RMB 27 billion in the no-policy scenario. The welfare loss from the no-policy scenario would be greater, at RMB 216 billion, if we take into account the social cost of carbon.³⁵

Table 6: No-policy Effects

Δ Delivery price (RMB/Ton)	-161 [-176, -147]
Δ Domestic coal production (Million Tons/Year)	693 [627, 764]
Δ Coal consumption (Million Tons/Year)	642 [580, 711]
Δ Economic Surplus and Pollution Damages (RMB bn.)	
Economic Surplus	300 [262, 343]
Consumer	607 [539, 683]
Producer	-293 [-323, -264]
Resource Tax	-15 [-16, -13]
Avoided Pollution Damages	-327 [-351, -303]
Air Pollution	-235 [-250, -220]
Water Depletion	-92 [-102, -84]
Δ Welfare	-27 [-59, -0]

The table reports the average market outcomes and the 2.5%–97.5% range (square brackets) across simulations in 2017 at the estimated parameters. We simulate forward the equilibrium based on the shocks in the last month of 2016. We report the changes when the enforcement effect is removed, relative to the observed policy. Air-pollution damages are calculated from the simulated PM_{2.5} and PM₁₀ paths. Water depletion costs are valued at RMB 86.6 per ton of coal. Δ Welfare combines economic surplus with avoided pollution damages, excluding the social cost of carbon.

Allocative Inefficiency

We next quantify the policy’s inefficiency relative to an efficiently implemented quota. We note two types of inefficiencies. First, our estimates suggest that the policy raises infra-marginal costs and leads to within-province allocative inefficiency. Second, our estimates

³⁵The social cost of carbon is set at the income-adjusted level of 162 RMB per ton of CO₂, and one ton of consumed coal is assumed to generate 1.886 tons of CO₂. Water depletion costs are valued at RMB 86.6 per ton of coal.

indicate greater production restrictions in larger, lower-cost provinces such as Inner Mongolia, indicating cross-province inefficiency. Although cap-and-trade, or a productivity-based production restriction, is likely infeasible given the institutional constraints, understanding these inefficiencies not only sheds light on the welfare costs of the policy, but also enables us to apply the ranking criterion in Section 3 to compare the policy with a hypothetical tax.

We first decompose the inefficiencies caused by the supply function shift in each province. Holding the 2017 equilibrium quantity fixed in each simulation path, we re-compute producer surplus using the pre-policy supply curve integrated up to that quantity. The within-province inefficiency is therefore the increase in producer surplus under the pre-policy supply curve relative to the observed policy, holding fixed the same provincial quantities. This inefficiency is large, estimated to be RMB 156 billion.

Next, we measure the total inefficiency. An efficient quota allocation across provinces requires the marginal surplus loss from an additional unit of reduction to be equalized across provinces. Therefore, holding fixed a time series of simulated shocks in 2017, for any total reduction ΔQ , the efficient allocation is the one implied by a uniform tax that achieves the same total reduction. We use the following procedure to find the efficient allocation: in each simulation based on the post-policy supply function, we (i) solve for the tax that delivers the same aggregate reduction in equilibrium, (ii) fix the quantities produced in each province under this tax, and (iii) compute economic surplus by integrating demand and the pre-policy supply functions up to those efficient quantities. We estimate a total producer-surplus inefficiency of RMB 246 billion, so 63.3% of the producer-surplus inefficiency is within province.

Table 7: Tax Effects (RMB bn.) Relative to the Observed Policy

	(1) No Policy	(2) Optimal	(3) Optimal, Weighted
Δ Consumer Surplus	607	-496	-219
Δ Producer Surplus	-293	-748	-679
Δ Tax Revenue	-15	1301	1050
Δ Saved Pollution Damages	-327	209	101
ΔW	-27	266	253

The table reports changes in welfare components in 2017 relative to the observed policy. The three columns report the no-policy counterfactual, the tax that maximizes $CS + PS + \text{tax revenue} + \text{saved pollution damages}$, and the best tax when $w_{\text{damage}} = 1$ and $w_{PS} = 1.372$. Saved pollution damages include local air-pollution health benefits and water depletion, and exclude carbon. ΔW is the equal-weight sum of the four welfare-component changes in each column. The corresponding tax values are 0, 429, and 322 RMB/ton in columns (1) through (3), respectively. Results are based on the baseline model.

7.2 Is Tax More Efficient?

In Section 2.3.2, we discussed the political reasons why production restrictions, as opposed to a tax, are used to implement the reform. Yet, the economic question remains: is it more efficient to use a tax? Section 3 shows that whether the best tax generates a higher welfare than the best production restriction relies on the slopes of the demand, cost and marginal damages as well as their variances. We present two sets of results that point to the better efficiency of tax.

7.2.1 Optimal Tax Maximizing Economic Surplus and Health Benefits

We first consider a national government’s objective function that takes into account the impact on economic surplus and health from coal:

$$W = E(\text{consumer surplus} + \text{producer surplus} + \text{tax revenue} - \text{pollution damages}), \quad (18)$$

where the damages correspond with the monetary values of the environmental damages in Section 7.1, excluding the social cost of carbon. An optimal counterfactual tax is set at the beginning of 2017 and maximizes the objective above, where the expected value of each component in the function is based on the estimate AR(1) processes of demand and marginal cost shocks and the realized values in the last month of 2016.

We find an optimal tax of RMB 429/ton, and column (2) of Table 7 shows the difference relative to the observed policy. Under this tax, the losses in consumer and producer surpluses are more than offset by tax revenues and additional saved pollution damages. Absent the hurdles to redistribute the tax revenues (Section 2), consumers and producers can both be made better off.

We also note that the optimal tax increases welfare by RMB 266 billion relative to the observed policy, whereas removing the policy altogether reduces welfare by RMB 27 billion. This gain arises because the observed equilibrium is on relatively inelastic portions of both demand and supply, allowing taxation to generate substantial revenue before higher tax rates induce more elastic responses from either side of the market.

Anticipation of the Demand and Supply Shocks An assumption in the calculation above is that the unobservables in the demand and supply functions are also unknown to the market participants or policymakers. This assumption is important for correctly defining expected welfare, but our qualitative results on the efficiency of the tax are likely robust to weaker information assumptions. In the extreme case where the unobservables are fully known to buyers, sellers and the government, a tax would generate less allocative inefficiency and

would be unambiguously more efficient than a production restriction for the same quantity target.

7.2.2 Optimal Tax with a General Loss Function

Next, we show that the inefficiency result can be significantly strengthened. A government objective function may take into account broader effects of coal consumption such as carbon emissions, mine-site ecological damage and mining accidents. A reasonable sufficient statistic for these consideration is the total coal output. In China’s case, the 5-year planning document in 2016 (NDRC, 2016*a*) explicitly calls for an annual production target of 3.9 billion tons of coal by 2020. Therefore, a more general objective function for the government can be augmented from W in (18) as

$$\begin{aligned} W^* &= E(\text{consumer surplus} + \text{producer surplus} + \text{tax revenue} - \text{pollution damages} - L(Q)) \\ &= W - E(L(Q)), \end{aligned}$$

where L is an increasing function of the total quantity Q .³⁶

We show that the optimal tax in the previous section achieves a higher W^* than the observed policy, despite not knowing the exact functional form of L . The method draws on a celebrated result in the decision theory (Quirk and Saposnik, 1962; Hadar and Russell, 1969): for an increasing function L , given two random variables Q_1 and Q_2 , $E(L(Q_1)) \leq E(L(Q_2))$ if the random variables’ distributions satisfy $F_1 \geq F_2$ (first-order stochastic dominance).

This result allows us to compare $E(L)$ between the optimal tax and the observed policy. Specifically, if the distribution of total output Q_τ under the tax is stochastically dominated by the distribution under the production restriction, then the tax’s $E(L)$ must be lower. Given that the tax achieves a higher value of W , it follows that W^* must be higher under the tax. Intuitively, if the quantity under the tax is stochastically dominated, the average quantity and the average pollution damages must be lower.

Figure 4 shows that the quantity under the tax is indeed stochastically dominated by that under the policy.³⁷ Moreover, a range of taxes around the optimal tax can also achieve

³⁶Leakage is an externality that we do not explicitly account for. Next to coal (65%), hydro (20%) had the second largest share in the power generation mix in 2016, followed by wind (4%) and nuclear (4%). Natural gas accounted for 3% of the generation mix (Göß and Niggemeier, 2017). In the long run, reduced coal consumption, especially in electricity generation, is likely to be offset by both the growth of renewable generations and increased natural gas use (EIA, 2022). Generations from these sources typically have lower local pollution intensities and carbon emissions than coal. Therefore, such substitution would still be expected to reduce the overall external loss when coal consumption falls.

³⁷There are several ways to formally test the stochastic dominance. First, for a simple but formal statistical test, we test an even stronger condition, which is that the quantity under the optimal tax based on the estimated parameters is statistically smaller than under the policy for each simulated path of shocks. More

higher W with stochastically dominated quantity distributions.

A key intuition for why such taxes exist is that the observed production restriction is an inefficient way to reduce output. By constraining low-cost producers, the policy makes the aggregate marginal cost curve steeper. A tax reallocates production toward lower-cost firms, allowing the same level of economic surplus to be achieved with lower aggregate output and therefore lower damages under any increasing loss function.

This result goes beyond the local comparison in Section 3. The local approximation assesses how a tax and a production restriction differ in their response to shocks near a common benchmark quantity $\hat{q}$. In the analysis here, the restriction rotates a significant portion of the supply curve and misallocates production across provinces, creating large inframarginal losses. These losses are not fully captured by the local approximation and create additional room for a tax to preserve surplus while shifting the entire distribution of output downward.

Finally, the same logic can be interpreted in a “dual” way to compare policies when marginal pollution damages are relatively well measured but the economic-surplus effects of regulation are not. Suppose a pollution-control policy reduces market quantity below the no-policy equilibrium. Under the conditions that private surplus increases and cost decreases in quantity, economic surplus is then increasing in quantity up to the no-policy equilibrium level. Therefore, between two policies that reduce in pollution damages by the same amount, the policy that induces a stochastically dominating distribution of market quantity, thus preserving higher output, delivers higher welfare.

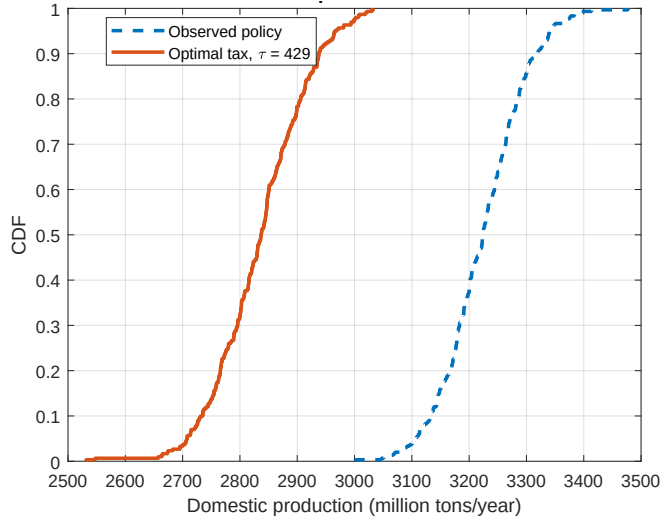
7.3 Policy Preferences

The results above show that the government was willing to forgo over RMB 266 billion of economic surplus and saved pollution damages when choosing the production restriction over a tax. As Section 2 and Section 3 show, policy priorities of improving producer surplus and reducing pollution damages may explain this choice. We therefore consider the following weighted policy function

$$\begin{aligned}\tilde{W}(w_{\text{PS}}, w_{\text{damage}}) = & E(\text{consumer surplus} + w_{\text{PS}} \cdot \text{producer surplus} \\ & + \text{tax revenue} - w_{\text{damage}} \cdot \text{pollution damages}),\end{aligned}$$

generally, given sufficiently many simulation draws, we can ignore the simulation errors and compute the maximal difference between the two CDFs over their support. The 95% confidence interval of this difference can be simulated using the asymptotic distribution of the estimated parameters.

Figure 4: Quantity CDFs: Production Restriction and Optimal Tax



We plot the CDFs of 2017 production under the observed policy and the optimal tax of RMB 429/ton.

A higher w_{PS} makes the production restriction policy more appealing, whereas the effect of a higher w_{damage} is ambiguous. We therefore consider a range of weight pairs, $(w_{\text{PS}}, w_{\text{damage}})$. For each pair, we solve for the optimal tax and then evaluate whether welfare is higher under that tax than under the observed production restriction policy.

We find that, for a given value of w_{damage} , the production restriction generates a higher W than the optimal tax when w_{PS} exceeds a threshold. Table 8 reports these thresholds. When the government places the same weight on pollution damages as consumer surplus ($w_{\text{damage}} = 1$), over-valuing the producer surplus by 37% is sufficient for the observed policy to dominate any tax. Column (3) of Table 7 shows that the best tax given such policy weights, RMB 322/ton, would generate RMB 253 billion of gains in the unweighted surplus W , but would be equal to the observed policy in the weighted policy objective value $\tilde{W}$.

This result reflects several important realities in China’s coal industry.³⁸ First, the profits of coal mining firms, many of which are state-owned, are not entirely private rents. Part of the state owned firms’ after-tax profits flows back to the government. This gives the government a direct fiscal stake in coal-firm profitability. Second, the surplus helps to stabilize employment and avoid worker resettlement. The falling prices in 2014-2015 forced provincial governments to reduce local taxes and expand safety net (Shanxi, 2015, 2016). Third, the profits reduce the need for external financing when the government encourages investment in efficient upgrades (Yuan, 2025). The government chooses the production restriction policy

³⁸This result is different from some policy preference estimates in other settings. For example, Lim and Yurukoglu (2018) estimates that government places much higher weight on consumer surplus than on utilities in the utility rate-setting context in the US.

despite the significant increase in tax revenues under the more optimal tax, which reflects the significant hurdles for redistributing the tax revenues (Section 2).

Table 8: Policy Weights That Rationalize Production Restrictions Over Taxes

w_{damage}	1.00	1.25	1.50	1.75	2.00
$w_{\text{PS}} \geq$	1.372	1.426	1.506	1.605	1.722

Notes: The table reports the range of w_{PS} given w_{damage} such that the observed policy achieves a higher policy objective value than the best tax. The tax values that correspond with the minimal w_{PS} values are 325, 425, 525, 600, 675 RMB/ton for $w_{\text{damage}} = 1, 1.25, 1.5, 1.75$, and 2, respectively.

We also note that a higher w_{damage} alone does not make production restriction more attractive. This result is related to our finding in Section 7.2.2: given our demand and supply estimates, a tax can often achieve higher economic surplus at lower quantities, thus lower pollution damages, compared with the production restriction policy. Therefore, tax becomes a more attractive instrument at higher values of w_{damage} , and a higher threshold for w_{PS} is required to rationalize production restriction against a tax.

8 Conclusion

We study China’s 2016 coal industry reform, which restricted coal production through mine closures and output controls. Empirically, we find that the policy reduced coal output, raised prices, and shifted surplus from consumers to producers. Using an estimated equilibrium model of provincial coal demand and supply, we quantify these distributional effects and compare the observed production restriction with alternative instruments.

Our welfare analysis shows that the production restriction was not the surplus-maximizing way to reduce coal consumption under equal welfare weights. A tax could have reduced aggregate output while raising total welfare, even after accounting for environmental and health benefits from avoided coal use. This finding highlights the efficiency cost of using production restrictions rather than price-based instruments. At the same time, implementing such a tax would have required substantial fiscal and institutional reform, whereas production restrictions could be imposed more easily.

We further show that the observed policy can also be rationalized by policy preferences. The reform documents emphasized not only pollution reduction, but also coal-industry profitability and economic stability in producing regions. Consistent with these objectives, we show that a tax may no longer dominate the observed policy when the government places sufficiently high weight on producer surplus. More broadly, we show that the ranking of taxes and production restrictions depends not only on demand, cost, and externality shocks,

but also on the weights attached to different components of policy objective function. We provide an empirical framework to bound these weights.

The results also underscore the challenges of transitioning to a low-carbon economy given the institutional constraints. Consistent with this observation, China has made massive investments in renewable energy. In 2024, for the first time, the share of coal in electricity generation fell below 60%, primarily driven by the growth of low-cost electricity production from renewable sources, as opposed to reduced coal consumption (Howe, 2024).

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A Policy Documents

Figure 5 provides a snapshot of the original government document for the supply reduction policy. We provide an English translation of this document in this section.

In order to implement the “*Opinions of the State Council on Resolving Overcapacity in the Coal Industry and Achieving Sustainable Development*” (Guo Fa [2016] No. 7, hereinafter referred to as the “Opinions”), and to further regulate and improve the coal production and management order, effectively resolve overcapacity, and promote the sustainable development of coal enterprises, the following notice is issued regarding related issues:

1. Fully understand the importance of standardizing and improving the order of coal production and operation

Currently, the coal industry is in a serious predicament. Especially since last year, some coal enterprises have resorted to measures such as overcapacity production, compensating for price through volume, and low-price dumping to maintain production operations in an effort to seize market share. These actions have not only severely disrupted the normal order of coal production and operation but have also exacerbated the imbalance between supply and demand in the market, intensifying the difficulties faced by the industry. If these issues are not promptly curbed, the national goals and tasks related to the development of the coal industry will be difficult to achieve. All regions and enterprises must adopt a strategic perspective that promotes supply-side structural reform and effectively addresses capacity reduction tasks. It is essential to enhance overall awareness and fully recognize the importance of standardizing and improving the order of coal production and operation for facilitating the recovery and development of the coal industry. This requires unifying thought and action with the spirit of the “Opinions”, taking effective measures to ensure the implementation of regulations concerning the control of overcapacity production, reduction of production volume, suspension of production during holidays, and maintenance of fair competition.

2. Main Measures to Standardize and Improve Production and Operation Order

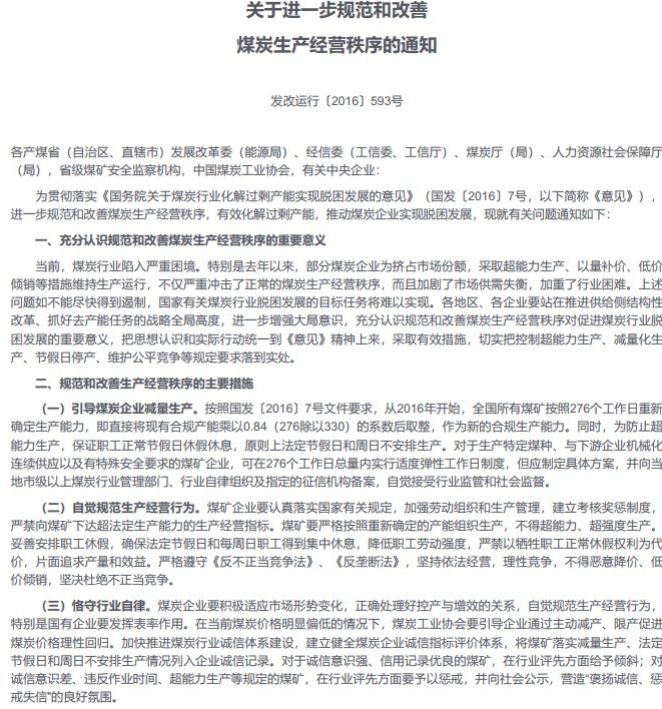
- (a) **Guiding Coal Enterprises to Reduce Production Volume:** In accordance with the requirements of the State Council Document [2016] No. 7, starting from 2016, all coal mines nationwide are to re-determine their production capacity based on 276 working days. This means directly multiplying the existing compliant capacity by a coefficient of 0.84 (276 divided by 330) and rounding to the nearest whole number to establish the new compliant production capacity. Additionally, to prevent overcapacity production and ensure employees can take their normal holidays,

no production should be scheduled on statutory holidays and Sundays. For coal mines producing specific coal types, providing mechanized continuous supply to downstream enterprises, or with special safety requirements, a moderately flexible working day system may be implemented within the total of 276 working days, provided that a specific plan is developed and filed with local coal industry management departments at or above the municipal level, industry self-regulatory organizations, and designated credit institutions, and that these enterprises consciously accept industry regulation and social supervision.

- (b) **Consciously Standardizing Production and Operation Behavior:** Coal mines must earnestly implement national regulations, strengthen labor organization and production management, and establish a performance assessment and reward-punishment system. It is strictly prohibited to set production and operation targets that exceed the legal production capacity for coal mines. Production must be organized strictly according to the re-determined capacity, and overcapacity or excessively intense production is not allowed. Employee leave must be properly arranged to ensure that staff can rest during statutory holidays and every Sunday, thereby reducing labor intensity. It is forbidden to sacrifice employees' normal leave rights in order to pursue output and efficiency. Coal mines must strictly adhere to the "Anti-Unfair Competition Law" and the "Anti-Monopoly Law," operate in accordance with the law, engage in rational competition, refrain from malicious price cuts, and eliminate unfair competition.
- (c) **Upholding Industry Self-Regulation:** Coal enterprises must actively adapt to changes in market conditions, correctly balance production control and efficiency enhancement, and consciously regulate their production and operation behavior, especially state-owned enterprises which should set a leading example. In the current context of notably low coal prices, the coal industry association should guide enterprises to promote rational price recovery through proactive production reduction and output limitations. The construction of an integrity system in the coal industry should be accelerated, establishing and improving an integrity index evaluation system for coal enterprises. Compliance with production reduction, as well as not scheduling production on statutory holidays and Sundays, should be included in the integrity records of enterprises. Coal mines with strong integrity awareness and good credit records should receive favorable treatment in industry evaluations, while those with poor integrity awareness, violations of operational hours, and overcapacity production should face penalties in industry evaluations and be publicly disclosed to create a positive atmosphere of "rewarding integrity

and punishing dishonesty.”

Figure 5: Policy Document: Supply Reduction Policy



B Data Comparison

To validate the provincial and monthly production and consumption data we compiled from various sources, we compare the statistics based on our data and those from the National Bureau of Statistics of China in Table B.1. The first column displays the national values aggregated from our data based on coal transportation, the second column is derived from the NBS Coal Balance Table for all provinces, and the third column represents the national statistics in the NBS National Coal Balance Table.

Overall, the national statistics derived from our transportation data are reasonably close to those in NBS. For instance, both datasets show declining output and consumption between 2014 and 2016, and a decrease in imports in 2015 followed by a rebound in 2016.

We also validate our data at the province-year level. Figure B.1 compares the shipment out of and into a province in each year with the annual coal production and consumption data by province in the provincial coal balance tables from NBS. Overall, they align closely, with the imputed consumption data being slightly lower than those reported by the NBS. This difference is expected because NBS data explicitly state that they do not remove certain

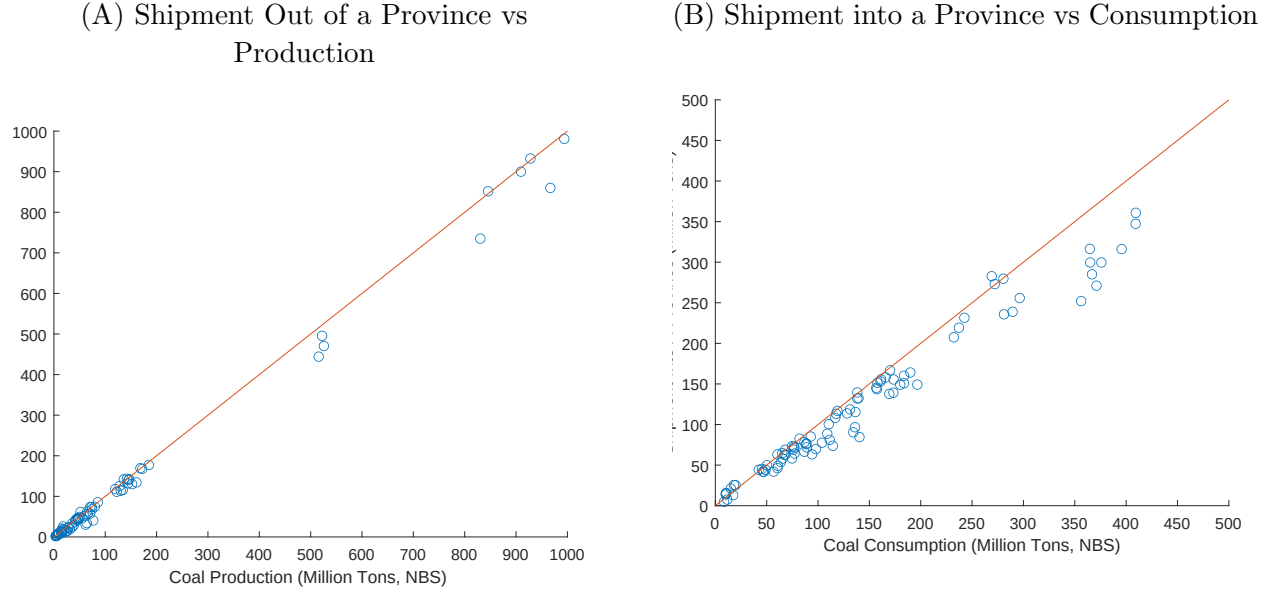
Table B.1: Data Comparison of Coal Output and Consumption (million tons)

	2014			2015			2016		
	Main	NBS (Prov)	NBS (Nat)	Main	NBS (Prov)	NBS (Nat)	Main	NBS (Prov)	NBS (Nat)
Output	3703	3876	3874	3457	3747	3747	3135	3411	3411
Consumption	3860	4317	4136	3662	4255	3998	3574	4249	3888
Thermal power	1974	1823	1895	1872	1792	1796	1908	1818	1827
Heating supply	181	306	224	204	323	241	222	345	266
Coking Coal	563	667	639	530	625	609	536	626	606

Notes: The table compares the transportation data used in the empirical analysis (under main columns) with the annual numbers reported by the National Bureau of Statistics of China. The column of NBS (Prov) contains data collected from province coal balances tables and aggregated to national numbers, and the column NBS (Nat) uses statistics reported in the national coal balance tables. The sum of coal consumption across provinces exceeds the national total because of the potential double counting according to the official Q&A from China NBS (See https://www.stats.gov.cn/hd/cjwjtjd/202302/t20230207_1902276.html). Under the main column, the output is calculated as total shipment, and the monthly consumption is constructed in Appendix C. We then aggregate province-month production and consumption to the national level.

instances of double-counting at the province level.³⁹

Figure B.1: Data Comparison of Shipment, Coal Output and Consumption at Province Levels (Million Tons)



The figure compares province-year coal production and consumption in the data used in our empirical analysis with those reported by province coal balance tables. One dot in the figures represents one province-year pair. The y -axis is calculated by aggregating monthly total shipment out of and into a province within a year.

C Coal Consumption

We collect monthly coal consumption data from 2014 to 2016 across seven sectors: electricity, coking, construction, heating supply, steel-making, chemistry, and others. Table C.1 displays annual consumption in these sectors. Electricity is the largest coal consumer, accounting for over 50% of total consumption. Other major users include coking and cement plants, each accounting for 15%. The consumption breakdown is stable during our sample period. The “others” sector, which mainly includes residential demand and small-scale industrial processes, accounted for 7% in 2014 and declined to 3% in 2016. We plot the national time series of the downstream output in Figure C.1.

³⁹For example, coal transported to province A for washing and burned at power plants in B are both counted as consumption in provinces A and B. Our shipment data only count the final consumption in B.

Table C.1: Consumption Shares by Sectors

Year	2014	2015	2016
Electricity	0.51	0.51	0.53
Coking	0.15	0.14	0.15
Construction	0.15	0.15	0.14
Heating Supply	0.05	0.06	0.06
Steel-making (steam coal only)	0.04	0.04	0.04
Chemistry	0.04	0.04	0.04
Others	0.07	0.06	0.03

Note: The table shows annual coal consumption shares by sector. We collected monthly coal consumption by sector from China Coal Resource, and then aggregated them annually to calculate those shares.

To allocate coal consumption by sector across provinces, we further collect the monthly output of primary products in each sector by province from the NBS and convert it into coal consumption. For the electricity sector, we use thermal electricity output as the main product. Although thermal electricity generation includes both coal and natural gas, natural gas contributes to only about 4.5% of total thermal generation, making its impact negligible.⁴⁰ We use cement as the representative product for the construction sector, as it accounts for 83.7% of coal consumption there. In the chemistry sector, fertilizers serve as the primary product.

Provincial coal balance tables provide annual coal consumption for thermal electricity generation and coking. We aggregate thermal and coke output to the province-year level and calculate coal consumption per 1 MWh of electricity and per ton of coke for each province and year. We assume that these ratios remain constant across months within a province and year, converting thermal electricity generation and coke into coal consumption for the electricity and coking sectors. For cement, steel, and fertilizers, we aggregate their output by month across provinces and calculate the coal-to-output ratio. Assuming this ratio is constant across provinces within a month, we compute coal consumption in the construction, steel, and chemistry sectors based on their province-month output. Table C.2 shows that producing 1 MWh of electricity uses less coal over time, reflecting the gradual improvement in power plant energy efficiency. The conversion ratios are also consistent with engineering estimates by industry experts.⁴¹

⁴⁰<https://chinaenergyportal.org/en/2016-detailed-electricity-statistics-updated/>

⁴¹For example, producing 1 ton of cement requires 0.11 ton of standard coal, equivalent to a conversion ratio of 0.15 between coal and cement, given that 1 ton of raw coal equals 0.7143 ton of standard coal. See <https://www.chinanews.com.cn/cj/2011/05-03/3011403.shtml>.

Table C.2: Coal-to-End Product Output Ratio by Sectors

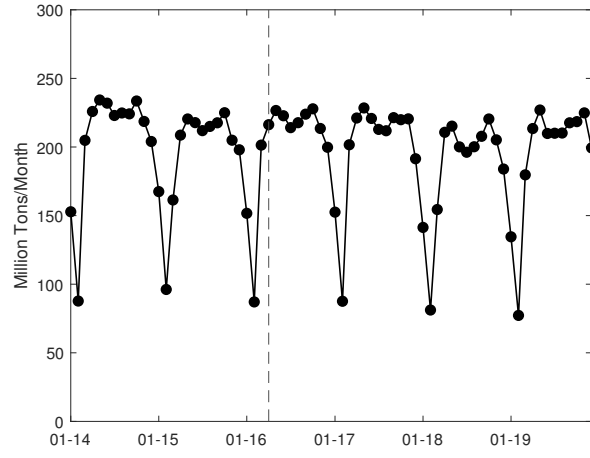
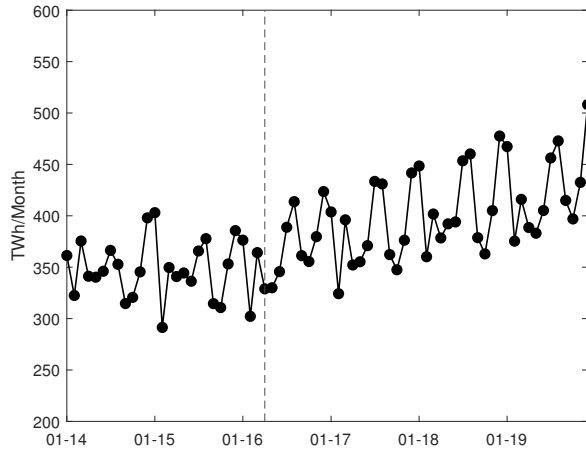
Year	2014	2015	2016
Electricity (Ton of coal/MWh)	0.47	0.44	0.43
Cement (Ton of coal/Ton)	0.23	0.24	0.21
Steel (Ton of coal/Ton)	0.14	0.12	0.13
Coke (Ton of coal/Ton)	1.18	1.19	1.20
Fertilizer (Ton of coal/Ton)	2.01	1.85	2.00

Note: The table reports the conversion ratios used to estimate coal consumption for the production of key downstream products in each sector. Monthly data on the output of thermal electricity, cement, steel, coke, and fertilizers by province is collected from the NBS. The provincial coal balance tables provide annual coal consumption for thermal electricity generation and coking. We aggregate thermal and coke output to the province-year level to calculate the coal consumption per 1 MWh of electricity and per ton of coke. The table shows the average across provinces for each year. For cement, steel, and fertilizers, we calculate national output annually and match it with coal consumption data for the construction, steel (steam coal only), and chemical sectors from China Coal Resource, resulting in an annual conversion rate applied uniformly across all provinces.

For the heating sector and the “other” sector, which lack representative products, we collect annual coal consumption for heating supply from provincial coal balance tables. We then compute annual consumption in the “other” sector as the residual of total coal consumption after accounting for electricity generation, heating supply, industry, and construction. Using these data, we first calculate province-specific annual coal consumption shares by sector and then allocate monthly consumption across sectors according to these shares.

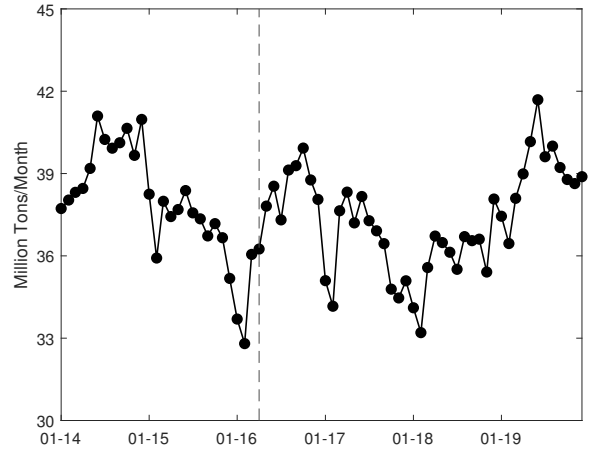
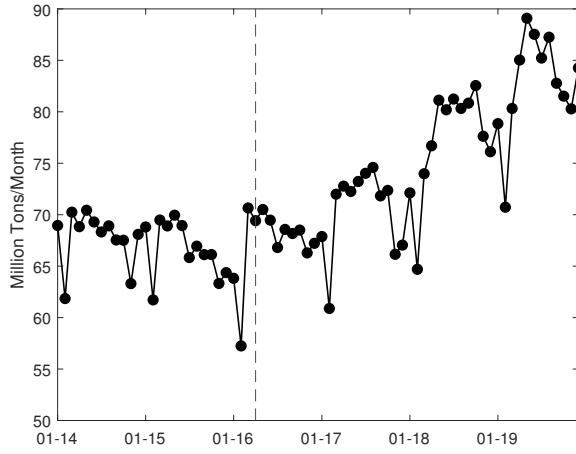
We extrapolate these data to 2017. Our monthly, sector-level consumption data and cross-province shipment data end in 2016. For 2017, we impute the consumption data using the province-level monthly production data from NBS. Specifically, we allocate each province’s production to destination provinces using shipment shares observed in 2016. We note that the shipment quantity is slightly lower than NBS’s production data in prior years, likely due to a double-counting issue with the NBS data as discussed in Appendix B. Therefore, the allocated production is scaled down by 0.936, the average production-to-shipment ratio observed in 2014-15. We assume that the total production allocated to a province is its consumption, which is a reasonable assumption given that coal inventory has stabilized in 2017 (Figure 1).

Figure C.1: Output in Major Downstream Sectors
(a): Thermal Electricity (b): Cement

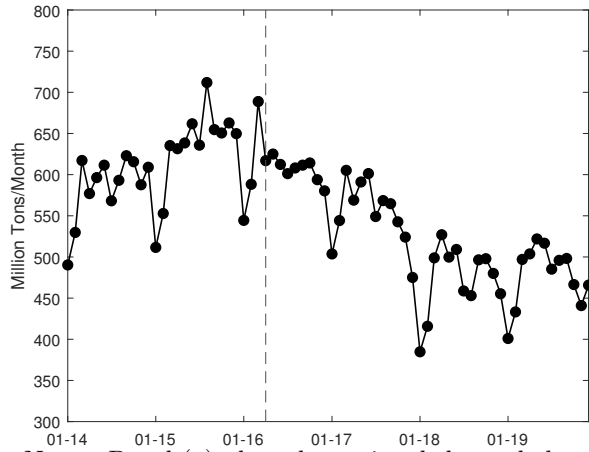


(c): Steel

(d): Coke



(e): Fertilizer



Notes: Panel (a) plots the national thermal electricity generation, (b) plots the national cement production, (c) plots the national steel production, (d) plots the national coke production, (e) plots the national fertilizer production.

D China's Major Infrastructure Development in April and May of 2016

The list below represents our best-effort search for major active infrastructure projects (with a cost over RMB 50 billion) in April and May of 2016, based on NDRC announcements, Xinhua News, China Daily and other news sources. We focus on projects that are extremely material-intensive, thus requiring significant use of coal and coke for concrete and steel. We exclude the Guangzhou–Shenzhen–Hong Kong Express Rail Link (Hong Kong section), which meets the RMB 50 billion cost threshold, but the scale (26km of railway) was far smaller than the other projects.

New Projects

1. Yuxi–Mohan Railway (China-Laos Railway) in Yunnan (southwest of China). Full-scale construction began on April 19, 2016, following earlier preparatory works that started in September 2015. The project's total cost is about RMB 51.6 billion.⁴²
2. Beijing–Zhangjiakou High-Speed Railway (HSR) in Beijing and Hebei (central China). The construction started on April 29th, 2016 (Wang, 2021), with a total investment of RMB 58.4 billion.⁴³
3. Tianfu International Airport in Chengdu (southwest of China). The construction started in May 2016, with a total investment of RMB 72 billion.⁴⁴

Ongoing Projects

1. Haoji Railway between Shanxi and Inner Mongolia (northwest-central China). The construction started in 2015 and was completed in 2019, with a cost more than RMB 193 billion.⁴⁵
2. Daxing International Airport in Beijing and Hebei (central China). The construction started in 2014 and was completed in 2019, with a cost of RMB 80 billion.⁴⁶

⁴²<https://global.chinadaily.com.cn/a/202007/03/WS5efe84baa310834817256ef1.html>

⁴³<https://www.reuters.com/article/sports/china-approves-9-2-billion-winter-olympics-high-speed-rail-line-local-government-idUSKCN0SS04F/>

⁴⁴<https://aviationweek.com/air-transport/chinas-new-chengdu-airport-facilitates-massive-air-travel-growth>

⁴⁵<https://global.chinadaily.com.cn/a/201910/05/WS5d97d87fa310cf3e3556ecde.html>

⁴⁶https://www.chinadaily.com.cn/business/2014-12/26/content_19177195.htm

3. Beijing–Shenyang HSR in Beijing, Hebei and Liaoning (central-north China). The construction began in 2014 and was completed in 2021, with a cost of RMB 123.5 billion.⁴⁷
4. Hong Kong–Zhuhai–Macau Bridge in Guangdong, Hong Kong and Macau (south China). The construction began in 2009 and was completed in 2018, with a cost of RMB 127 billion.⁴⁸
5. Wudongde Hydropower Station in Sichuan and Yunnan (southwest of China). The construction started in 2015 and was completed in 2021, with a cost of RMB 120 billion.⁴⁹
6. Lianghekou Hydropower Station in Sichuan (southwest of China). The construction started in 2014 and was completed in 2022, with a cost of RMB 66.5 billion.⁵⁰

E Coal Transportation

An integrated railroad and coastal marine shipping system is the primary mode of coal transportation in China, accounting for 65.2% of total coal transport in 2017.⁵¹ The transportation costs typically include the direct expenses associated with freight charges, which are influenced by the distance to be traveled, the mode of transportation (such as rail, truck, or barge), and the specific logistics involved. While shipping fares for waterway transport are influenced by market forces, railroad transportation fares are typically regulated.⁵² Additionally, loading and unloading fees, terminal charges, and any necessary handling costs contribute to the total transportation expenditure. Other factors, such as fuel prices, maintenance of transportation vehicles, and labor costs, also play a significant role in shaping these costs.

Due to the limited variation in transportation shares, directly estimating transportation costs through shipping choices via a behavioral model is challenging. Instead, we manually collected estimated transportation costs between major coal producers and consumers from an authoritative industry report (Cao and Feng, 2017) published by China Bond Rating Co.,

⁴⁷https://usa.chinadaily.com.cn/china/2013-12/30/content_17205381.htm

⁴⁸https://web.archive.org/web/20190327195326/http://www.xinhuanet.com/english/2018-10/23/c_137553194.htm

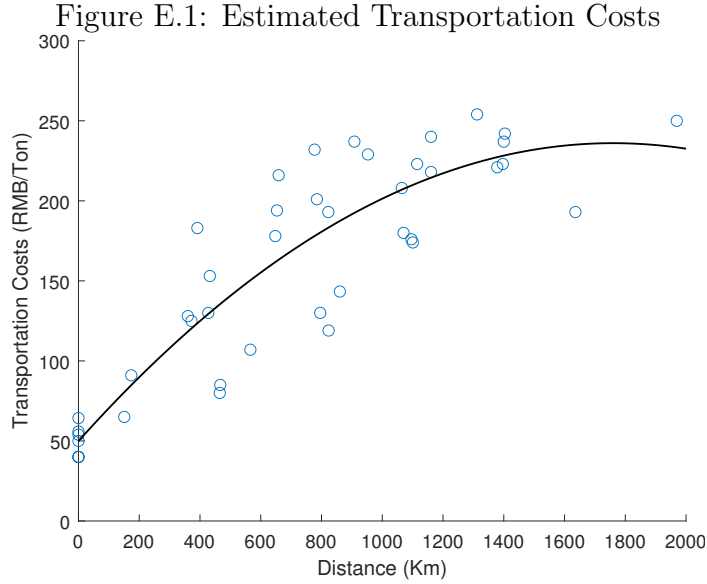
⁴⁹<https://www.reuters.com/business/energy/chinas-giant-wudongde-hydro-project-put-into-full-operation-2021-06-16/>

⁵⁰<https://www.chinadailyhk.com/hk/article/264174>

⁵¹See http://pdf.dfcfw.com/pdf/H3_AP201811151246522585_1.PDF.

⁵²For more details about the composition of regulated fares for transporting coal by railroad, please refer to <http://www.gov.cn/xinwen/2017-12/26/5250421/files/5a71188c4fd34d1d9f05ef5e16690d2e.pdf>.

Ltd., a leading agency that investigates and collects industry information to assess default risks of bonds across various sectors, including coal. Then we fit a quadratic relationship between the transportation costs and the distance between any two provinces, imposing the restriction that transportation costs increase with distance up to the maximum distance between any two provinces in China. Figure E.1 shows the transportation costs of province pairs estimated by the industry report, along with the estimated relationship between inter-provincial distance and transportation costs. The intercept represents the costs of transporting coal within a province, estimated to be RMB 60.9/ton. The estimates also indicate slight distance discounts.



Notes: The figure shows transportation costs (RMB/ton) between major coal-producing and coal-consuming provinces, plotted against the distance between their capital cities (km), based on an industry report (Cao and Feng, 2017) published by China Bond Rating Co., Ltd. We then approximate the relationship between transportation costs and distance using a quadratic function (black curve).

F Simulation of Demand and Supply Shocks

We estimate the following AR(1) processes for demand shocks ε_{jt} and supply shocks ξ_{it} :

$$\varepsilon_{jt} = \delta_0^D + \delta_1^D \varepsilon_{jt-1} + \omega_{jt}^D$$

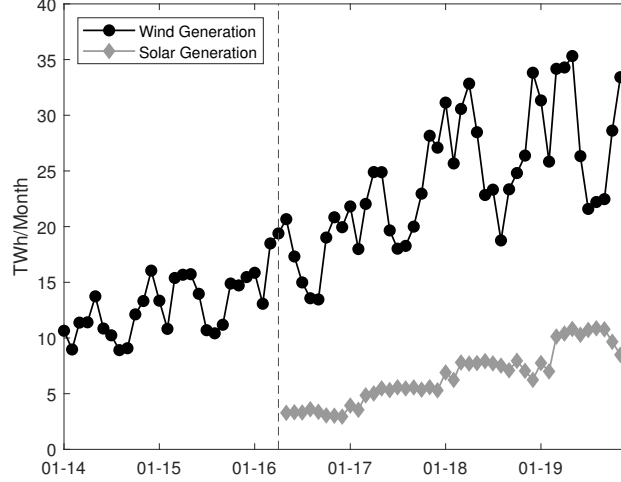
$$\xi_{it} = \delta_0^S + \delta_1^S \xi_{it-1} + \omega_{it}^S,$$

Table F.1 reports the estimated AR(1) coefficients, showing that demand shocks exhibit greater persistence than cost shocks.

	Demand	Domestic Supply
δ_1	0.805 (0.025)	0.494 (0.066)
N	1020	850

Table F.1: Estimated AR(1) Process for Demand and Cost Shocks

Figure F.1: Wind and Solar Generation in China



Note: We plot the total monthly wind and solar generation in China. The solar generation data are available after April 2016, and their levels are considerably lower than the wind's.

To forecast future demand, we additionally model the evolution of wind generation and cooling degree days (CDD). Motivated by the growth pattern of the total wind generation at the national (Figure F.1) and province level, we model the generation as a province-specific linear time trend with province-month-of-the-year fixed effects and shocks:

$$X_{jt}^{\text{Wind}} = \delta_{jm(t)}^{\text{Wind}} + \delta_{j1}^{\text{Wind}} t + \omega_{jt}^{\text{Wind}},$$

which is estimated separately for each province. We assume that $\omega_{jt}^D, \omega_{it}^S, \omega_{jt}^{\text{Wind}}$ are mutually independent in the forward simulation. We re-sample historical CDDs by year, under the assumption that cross-year climate variation over the sample period is stable.

G Long-Term Contracts and Demand Elasticities

An important assumption in Section 5 is that the monthly coal price in a province is determined in the market equilibrium. In this section, we explore the robustness to this assumption by examining the sensitivity of demand elasticities when a share of demand is locked in by

the contracts.

Specifically, we assume that long-term contracts, often signed at the beginning of a year, commit buyers and sellers to trading a fixed quantity at a fixed price for the coming year. This is the original intent of the long-term contracts, although, as Section 2 shows, few buyers and sellers were compliant in 2014 and 2016. To see how the potential contractual lock-in affects estimated demand elasticities, we consider the following robustness analyses: we assume that a share of $x_{jn}\%$ of observed quantity demanded in year n and province j , Q_{jn}^* , is the contracted amount. This amount is purchased in equal amount over the 12 months. Therefore, the coal demand in the spot market in Equation (10) becomes

$$\ln(q_{jt}(p_{jt}) - \frac{Q_{jn}^*}{12}x\%) = \alpha^D \ln p_{jt} + \beta_X X_{jt} + FE_j^{\text{demand}} + FE_{m(t)}^{\text{demand}} + \varepsilon_{jt} \quad (\text{G.1})$$

For simplicity, we assume that the share is uniform across provinces $x_{jn} \equiv x_n$ and differs across years. In Table G.2, we report the estimates of the spot market elasticity α^D and the quantity-weighted average adjusted elasticity based on the total quantity for two sequences of x_n . In panel A, we assume that the fixed quantity share is 7% for 2014-2016, and is 50% for 2017. These estimates are based on the discussion in Section 2.⁵³ In panel B, we exclude 2017, the year most affected by the contracted coal and re-estimate. We find that the implied elasticities in both panels are smaller, but still statistically not distinguishable from the baseline estimates in Section 5.

⁵³In reality, the locked-in amounts likely differ across months in a year, especially in 2014-2016, but it remains of interest whether our demand functional form is robust to the lock-in effects from such contracts.

Table G.2: Locked-Contract Demand: Quantity-Weighted Average Elasticities

<i>Panel A: 2014–2017</i>				
	Timing	SOE	Small-Mine Capacity	Timing+SOE
α^D	-0.703 (0.212)	-0.633 (0.201)	-1.309 (0.502)	-0.648 (0.202)
Adj. avg. elasticity	-0.590 (0.178)	-0.531 (0.169)	-1.099 (0.421)	-0.544 (0.170)
<i>Panel B: 2014–2016</i>				
	Timing	SOE	Small-Mine Capacity	Timing+SOE
α^D	-0.431 (0.192)	-0.395 (0.195)	-0.986 (0.402)	-0.408 (0.192)
Adj. avg. elasticity	-0.403 (0.180)	-0.369 (0.183)	-0.922 (0.376)	-0.381 (0.180)

Notes: The table reports locked-contract demand estimates for two samples. Panel A uses annual contract shares [0.07, 0.07, 0.07, 0.5] for 2014–2017. Panel B uses [0.07, 0.07, 0.07] for 2014–2016. Each estimate is from the log spot-demand regression using the indicated IV set. The adjusted average elasticity accounts for fixed quantities by scaling the spot-demand price coefficient by the spot-demand share in total demand for each province-month and then taking the quantity-weighted average across province-month observations. Standard errors for the adjusted average elasticity treat the fixed-quantity shares as fixed and apply the same quantity-weighting transformation to the coefficient standard error. All specifications exclude Beijing and include province and month fixed effects plus wind generation and cooling degree days.

H Coal Consumption and Air Pollution

To quantify the environmental impact of reducing coal production, we additionally model air pollution as a function of coal consumption and health benefits from changes in pollution levels.

H.1 Air Pollution

We specify a mapping from coal consumption to monthly PM2.5 and PM10 concentrations within a province,⁵⁴ which are the primary predictors of health costs in the literature we use later. We estimate the following model for the concentration of pollutant $\ell \in \{\text{PM2.5, PM10}\}$ in province j and month t (where the coefficients are ℓ -specific):

$$\ln \ell_{jt} = \psi_C \ln \tilde{C}_{jt} + \psi_X X_{jt} + F E_j^{\text{pollution}} + F E_{m(t)}^{\text{pollution}} + \omega_{jt}. \quad (\text{H.2})$$

⁵⁴We use the average of all monitoring stations' concentration readings in a province.

In the above, to account for spillover effects (Fu, Viard and Zhang, 2022), we define the variable $\tilde{C}_{jt}$ as the total coal consumption in province j and all of its neighboring provinces in period t .⁵⁵ The variable X_{jt} includes meteorological conditions, including temperature, dew point temperature, sea level pressure, wind speed, sky condition total coverage code, precipitation duration and depth. Additionally, we control for fixed effects at the province and month-of-the-year levels.

Coal consumption and the unobservable ω_{jt} may be correlated. Specifically, the coal consumption could be correlated with other economic activities that also contribute to pollution, but also correlated with heightened monitoring and regulation that reduces pollution. We therefore use the same IVs as the demand estimation for coal consumption.⁵⁶

Table H.3 indicates that a 1% increase in coal consumption in a province and its neighbors raises local PM2.5 concentration by about 1.4%. The effect is similar when we use PM10 concentration as the outcome variable.⁵⁷

Table H.3: The Effects of Coal Consumption on Air Pollutants

	(1) ln(PM2.5) OLS	(2) ln(PM10) OLS	(3) ln(PM2.5) IV	(4) ln(PM10) IV
$\ln \tilde{C}_{jt}$	0.416*** (0.115)	0.386*** (0.096)	1.438*** (0.332)	1.284*** (0.314)
Province FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
Meteorological Controls	YES	YES	YES	YES
Observations	1,440	1,440	1,440	1,440
Adjusted R ²	0.654	0.605	0.554	0.490
Kleibergen-Paap F Statistic			35.13	35.13

Notes: The table reports estimates of the effects of coal consumption on PM2.5 and PM10 concentrations by OLS and IV. All estimates control for meteorological conditions, including air temperature, dew point, sea level pressure, wind speed, sky condition coverage, and liquid precipitation duration and depth. The instruments used are the same as those in estimating demand. Standard errors are clustered at the province level.

⁵⁵We do not further normalize the variables by the province's size because both the pollution measure and the consumption are logged, and we control for province fixed effects.

⁵⁶We construct instruments for $\tilde{C}$ by summing the instruments across neighboring provinces.

⁵⁷Ito and Zhang (2020) estimate the elasticity of PM10 concentration with respect to coal usage to be 0.5 using a regression discontinuity design exploiting the Huai-River heating policy. This is similar to the OLS estimates from our province-month panel data, but smaller than the IV estimates.

H.2 Health Effects

We monetize morbidity and mortality costs associated with changes in ambient pollutant concentrations following Barwick et al. (2021).⁵⁸ Specifically, they estimate that a $10 \mu\text{g}/\text{m}^3$ reduction in daily PM2.5 generates \$9.2 billion in annual morbidity benefits. They also quantify mortality costs using a back-of-the-envelope calculation based on the mortality estimates in Ebenstein et al. (2017) (with data from 2004 to 2012). This calculation implies mortality costs of approximately \$13.4 billion per year associated with a $10 \mu\text{g}/\text{m}^3$ increase in PM10. We convert all estimates to a per-person per-month basis and express them in 2016 RMB.⁵⁹ In the simulation, we first predict the percentage changes of concentrations of both pollutants from coal consumption changes based on the marginal effects of coal consumption estimated using IV, and then translate the level changes to the total morbidity and mortality health costs per capita. The total changes are calculated by weighting the changes by the population for each province and then summing across provinces.

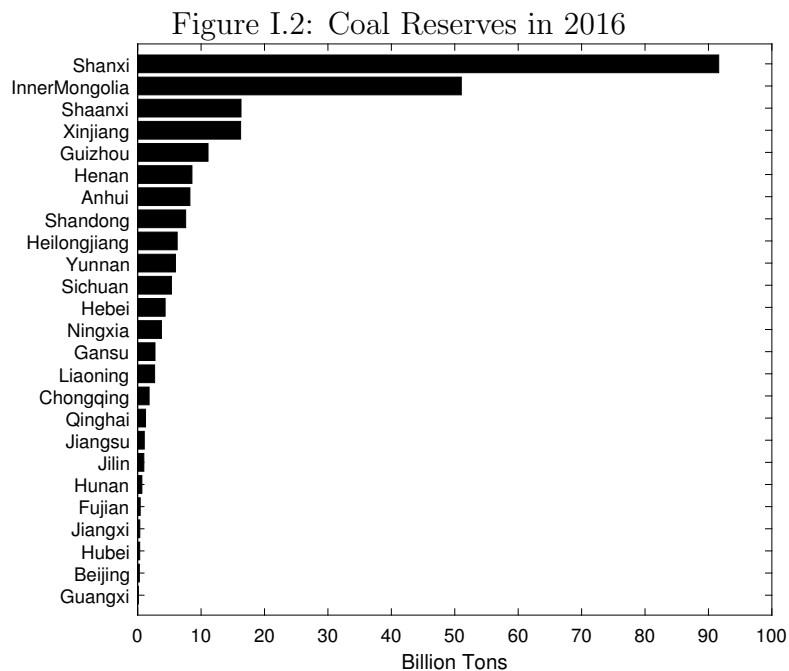
This approach likely understates the health benefits from reducing coal production by focusing solely on air pollution attributed to the resulting decrease in coal consumption. Other environmental damages associated with coal production (Chu et al., 2023), transportation, and storage of coal (Jha and Muller, 2018), such as air pollution from the excavation process, water contamination and land degradation, are not quantified here. We account for the environmental costs of water depletion at RMB 86.6 per ton of coal (Gu and Li, 2017). Nevertheless, addressing air pollution from coal consumption likely would have the greatest welfare impact given the large affected population and would be a high priority goal of the supply reduction policy (Feng et al., 2019).

⁵⁸An alternative is based on the approach in Ito and Zhang (2020) that estimate a household’s willingness-to-pay of 1.34 dollars per year for $1 \mu\text{g}/\text{m}^3$ of decrease in PM10 based on data from 2006 to 2014. They also derive an income-dependent willingness to pay based on their estimates of a random-coefficient logit model for air purifiers. This measure leads to a slightly lower monetary value for reducing air pollution.

⁵⁹The health cost estimates reported in Barwick et al. (2021) are expressed in 2015 U.S. dollars. We convert these estimates to RMB using an exchange rate of RMB 6.5 per USD. To obtain per-capita values and express them in real 2016 terms, we use a national population of 1.375 billion and consumer price indices (CPI) of 615.2 in 2015 and 637.5 in 2016. Finally, we divide the resulting annual per-capita values by 12 to obtain monthly measures.

I Additional Figures and Tables

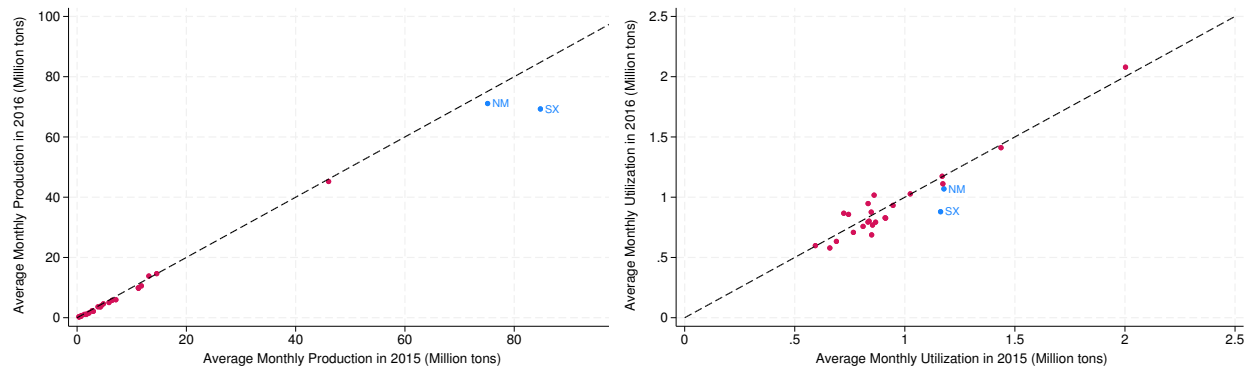
I.1 Distribution of Coal Reserves



Notes: Reserves are coal resources that have been surveyed and are feasible for excavation based on data from Qianzhan Industrial Research Institute.

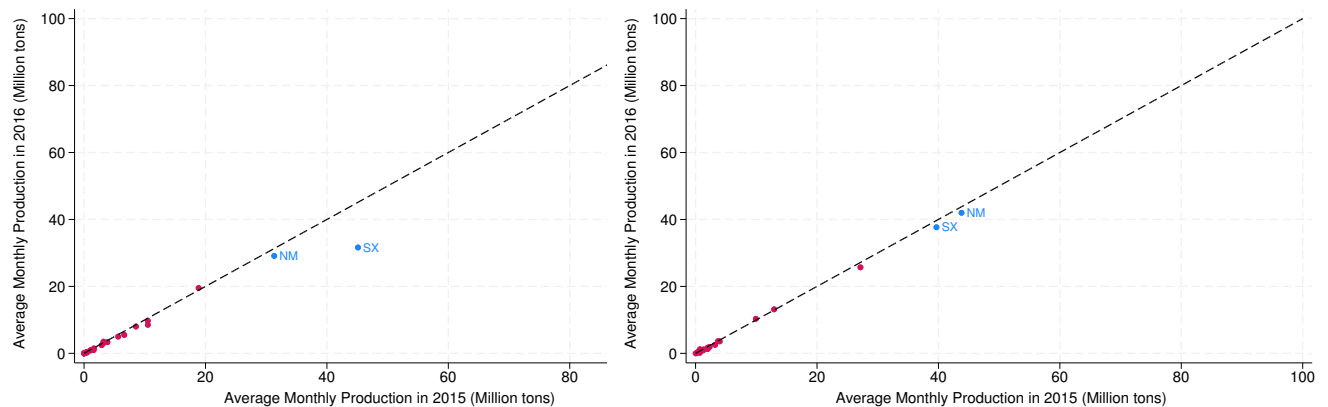
I.2 Policy Effects on Each Province and SOEs

Figure I.3: Shanxi and Inner Mongolia Reduce Production More in 2016
 (a): Year-to-Year Production (b): Year-to-Year Utilization Factor



Notes: In panel (a), each dot represents a combination of 2015 (horizontal) and 2016 (vertical) average monthly production of a province. A location below the 45 degree line indicates that the 2016 production is lower. In panel (b), each dot represents a combination of average monthly utilization factors, defined as production over capacity (both normalized to the monthly level), of a province. A location below the 45 degree line indicates that the 2016 utilization factor is lower.

Figure I.4: SOE Reduces Production More in 2016
 (a): Year-to-Year SOE Production (b): Year-to-Year Non-SOE Production



Notes: In panel (a), each dot represents a combination of 2015 (horizontal) and 2016 (vertical) average monthly production of a province's state-owned enterprises (SOE). A location below the 45 degree line indicates that the 2016 production is lower. In panel (b), each dot represents a combination of average monthly production by a province's non-SOE.

I.3 Concentration of Top State-owned Coal Companies

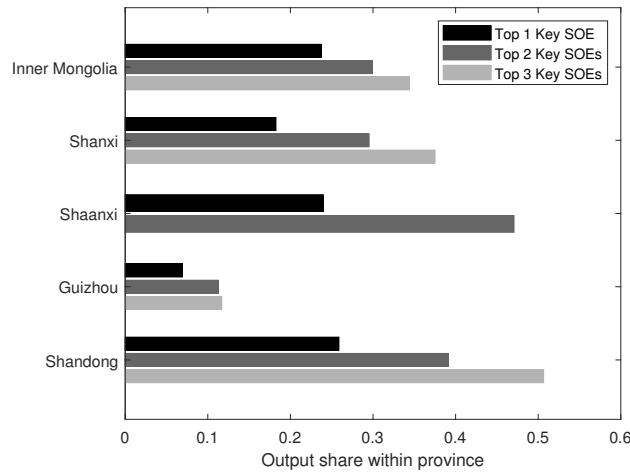
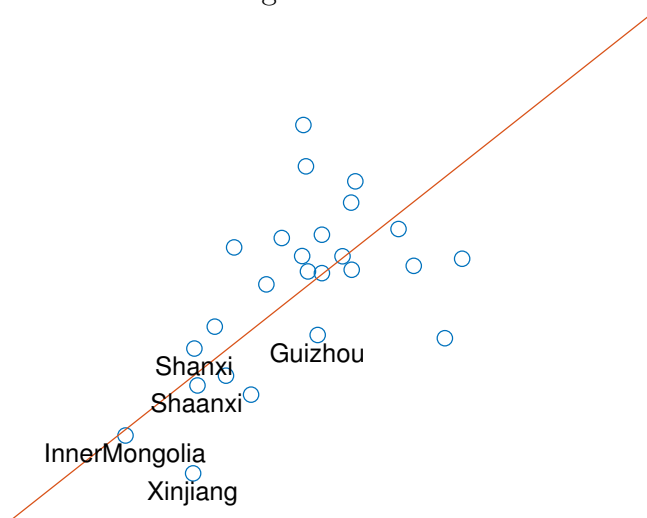


Figure I.5: Market Shares of Major SOE Coal Producers in Key Provinces

I.4 Comparison between Estimated Average Costs and Unit Costs from Financial Reports

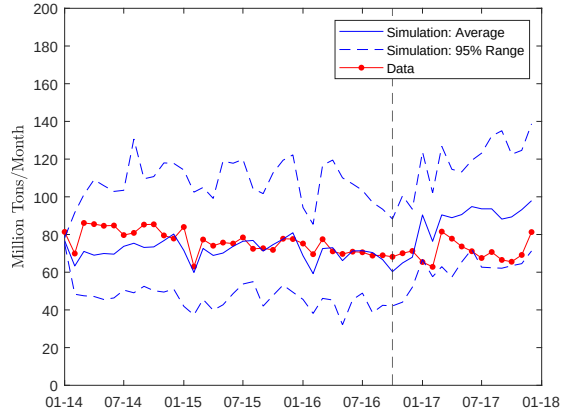
Figure I.6: Compare Estimated Average Costs with Unit Costs from Financial Reports



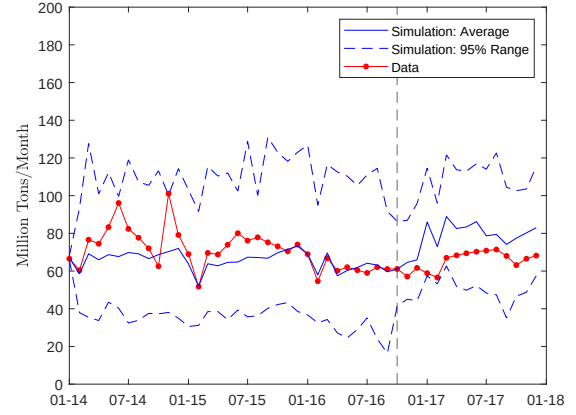
The figure compares average costs by province from the estimated model with unit costs from financial reports. The model's estimated average costs are calculated at the observed quantity and averaged between 2014 and 2016. The unit costs are collected from financial reports and bond prospectuses of major coal companies in each province, then averaged across all companies over three years in the sample.

Figure I.7: Simulations: Output by Provinces

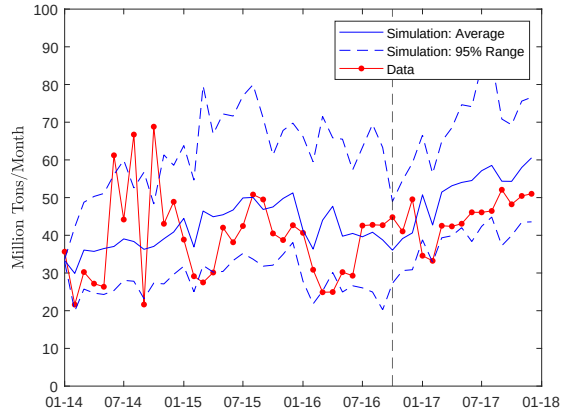
A: Inner Mongolia



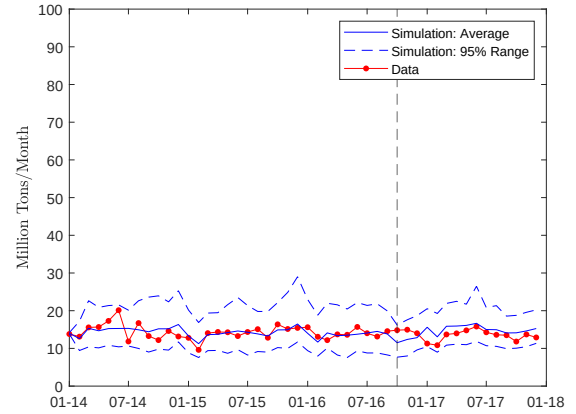
B: Shanxi



C: Shaanxi



D: Guizhou

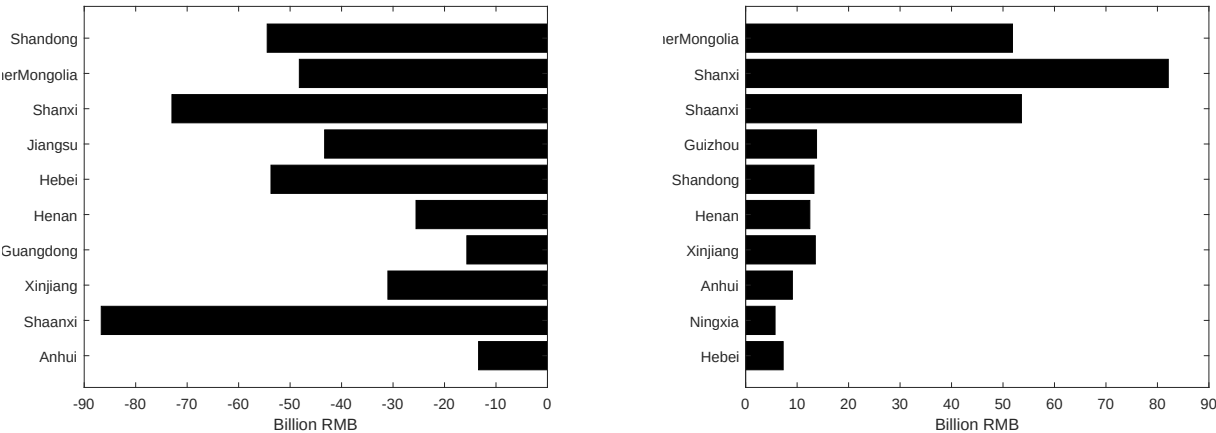


Figures plot the mean and 2.5%–97.5% range across all simulations for output in the four largest coal-producing provinces: Inner Mongolia, Shanxi, Shaanxi and Guizhou.

I.5 Model Fit by Provinces

I.6 Distribution of Economic Surplus by Top 10 Provinces

Figure I.8: Policy Effects on Economic Surpluses in Major Provinces
(a) Consumer Surplus (b) Producer Surplus



The figures illustrate changes in consumer surplus and producer surplus in 2017 relative to a no-policy counterfactual. Changes in consumer surplus are ordered by provinces' average pre-policy consumption, while changes in producer surplus are ordered by provinces' average pre-policy production.